DON BOSCO INSTITUTE OF TECHNOLOGY

(Autonomous Institute affiliated to VTU, Belgavi, Approved by AICTE, Recognised by the Government of Karnataka, NAAC A Grade Accredited).



DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

UNDERGRADUATE PROGRAM

(NBA Accredited 2025-2028)

OUTCOME BASED CURRICULUM & ASSESSMENT

Scheme and Syllabus

III & IV Semester

For the 2025 Admitted Batch

Kumbalgodu, Mysuru Road, Bengaluru 560074

Vision of the Institute

To be a center of excellence to transform young minds in technical and management education fostering innovation and entrepreneurial skills with ethical, environmental, and social responsibility.

Mission of the Institute

M1: To impart quality education in order to meet the needs of industry and society.

M2: To collaborate with academia, industry and research institutes to strengthen teaching and learning process.

M3: To promote equitable and harmonious development of students to work in teams.

M4: To imbibe lifelong learning skills and entrepreneurial skills exhibiting leadership.

Core Values

i) Attain excellence in different disciplines by creating, preserving and disseminating knowledge to all aspiring students

ii) Draw inspiration from the Institutions ethos and develop within its members a sense of accountability towards their community, society and the nation at large

iii) Accept the challenges globalization and changing times throw at us to offer high quality education and developmental services in a competitive manner

iv) Provide every opportunity to the Institutions key constituents–its faculty, staff, students and the community–to excel in their domain of expertise and contribute to every task with sincerity

v) Transition from the teacher – centric focus to the learner - centric approach in imparting knowledge

Vision of the Department
Empower the students to thrive for excellence in the area of information Science and Engineering to use technology for sustainable development for social needs

Mission of the Department
M1: To provide theoretical and practical Knowledge of Information Science and Engineering. M2: To provide inter disciplinary knowledge to be the leaders with social responsibility. M3: To collaborate with industry to facilitate content beyond syllabus. M4: To promote team work through projects. Co-curricular and social activities.

Knowledge and Attitude Profile (WK)	
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes	
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes	
PSO1	Apply the Mathematical tools, Electronics & Embedded Systems Knowledge and Programming knowledge to develop softwares.
PSO2	Use of Artificial Intelligence and Machine Learning. High performance computing, Cloud computing , Network security technologies and Software Engineering for providing solutions to the technological & social needs.
PSO3	Work individually and in teams, ethically exhibiting the managerial and leadership skills with sustainable Environment.

Program Educational Objectives	
PEO1	Graduates will contribute in development of software applications keeping abreast of the development.
PEO2	Graduates will exhibit competence as an individual and in teams with leadership and management skill.
PEO3	By optimizing the technology, the Graduates will be able to adopt lifelong learning..

COURSE CATEGORY

Ability Enhancement Courses	AEC
Basic Science Courses	BSC
Engineering Science Courses	ESC
Emerging Technology Courses	ETC
Humanities, Social Sciences, and Management Courses	HSMC
Internship	INT
Integrated Professional Core Courses	IPCC
Non Credit Mandatory Courses	NCMC
Open Elective Courses	OEC
Professional Core Courses	PCC
Professional Elective Courses	PEC
Project Work: Dissertation, Mini-project work and Major Project work	PROJ
Seminar	SEM

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4	B25CS34	Data Structures	15-18
5	B25IS35	Digital Design & Computer Organization	19-22
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7	B25CSL371	Data Analytics with EXCEL (Infosys certification course)	25-27
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B.E. in Information Science and Engineering
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SEMESTER: III

Sl No	Course Category	Course Code	Course Title	Teaching Dept. (TD)	Paper Setting Dept. (PSD)	Mapping to SDG#	Teaching Learning Hours					Examination			CREDITS
							L	T	P	TW & SL	TOTAL NLH	Duration in hours	CIE: SEE Marks	Total Marks	
1	ASC	B25MAT31A	Probability and Statistics	MAT	MAT	4,8,9,11, 17	42	28	0	50	120	03	50:50	100	4
2	IPCC	B25IS32	Object Oriented Programming using Java	IS	IS	4,8,9	42	0	28	50	120	03	50:50	100	4
3	IPCC	B25CI33	Operating Systems	IS	CI	4,9,11	42	0	28	50	120	03	50:50	100	4
4	PCC	B25CS34	Data Structures	IS	CS	4,9,11	42	0	0	48	90	03	50:50	100	3
5	PCC	B25IS35	Digital Design & Computer Organization	IS	IS	4,9,11	42	0	0	48	90	03	50:50	100	3
6	PCCL	B25CSL36	Data Structures Laboratory	IS	CS	4,9,11	0	0	28	02	30	03	50:50	100	1
7	AEC	B25xxL37x	Ability Enhancement Course - I	IS	-	4,9,11	0	0	28	02	30	03	50:50	100	1
8	SDC	B25ISSP38	Community Project (Project-Based Learning) / Societal Project	IS	IS	4,7,9,11	0	0	0	30	30	02	50:50	100	1
9	NCMC	B25NSK391	National Service Scheme (NSS)	NSS Coordinator	EC	3,4,5,6,13	0	0	0	28	28	-	100:0	100	-
		B25PEK391	Physical Education (PE) (Sports and Athletics)	PE Director	IS	3,5,8									
		B25YOK391	Yoga	Yoga Teacher	CI	3,4,13, 14,15,16									
		B25MUK391	Music	Music Teacher	AD	3,4,11									
		B25FAK391	Fine arts	Arts Teacher	EE	4,5,8,11, 12									

		B25SDG391	Sustainable Development Goals	IS	CS	1 to 17									
10	NCMC	B25DIP310	Additional Mathematics I	MAT	MAT	4,8,9	28	0	0	20	48	-	100:0	100	-
11	NCMC	B25PDK311	Personality Development	T&P	T&P	4	0	0	0	14	28	-	100:0	100	-
												TOTAL	**700:400**	**1100**	**21**

ASC: Applied Science Course, **PCC:** Professional Core Course, **IPCC:** Integrated Professional Core Course, **PCCL:** Professional Core Course laboratory, **AEC:** Ability Enhancement Course, **SDC:** Skill Development Course, **NCMC:** Non Credit Mandatory Course, **SDG:** Sustainable Development Goals, **L:** Lecture, **T:** Tutorial, **P:** Practical, **TW & SL:** Term Work and Self Learning, **NLH:** Notional Learning Hours, **CIE:** Continuous Internal Evaluation, **SEE:** Semester End Evaluation, , **PP:** Passed in Non Credit Course

Ability Enhancement Course - I

Course Code	Course Title	Mapping to SDG#	Course Code	Course Title	Mapping to SDG#
B25CSL371	Data Analytics with EXCEL (Infosys certification course)	4,9,11	B25ADL373	Project Management with GIT	4,9,11
B25ISL372	R Programming	4,9,11	B25CSL374	Unix Shell Scripting	4,9,11

Ability Enhancement Course Laboratory (AEC): An Ability Enhancement Course Laboratory is a practical, skill-oriented lab course designed to strengthen student's practical abilities, professional competencies that support communication, environmental awareness, computational thinking, interdisciplinary learning, and application skills through hands-on learning experiences. The laboratory may pertain to disciplinary or interdisciplinary involving experiments, design tasks, and mini-projects aligned with current industry practices.

Skill Development Courses (SDC): These courses are intended to develop specific practical skills and competencies that improve students' employability, technical proficiency, and professional readiness to bridge the gap between academic and industry requirements. These courses emphasize hands-on training, application of theoretical knowledge, and development of discipline-relevant and transferable skills required in industry and society, and develop entrepreneurship and start-up skills.

Community Project

A community is a social unit or group of people sharing socially-significant characteristics, such as place, set of norms, culture, religion, values, customs or identity. A community project involves addressing issues or needs within such a community or a network of entities working toward a common purpose. These projects may cover a wide range of areas, including welfare, sustainability, technology integration, and social development. Examples include establishing and maintaining an orphanage, implementing solar power generation and its maintenance, or developing environmental improvement solutions, etc. A community project is an experiential learning activity that encourages students to identify, analyse, and address real-life problems of the community using engineering knowledge. It aims to promote social responsibility and civic engagement, interdisciplinary thinking and collaboration and practical application of theoretical concepts, thereby enabling students to contribute meaningfully to community welfare and sustainable development. Students can take up project **individually or in a group not exceeding 4 students.**

The evaluation shall be done as per the following:

CIE: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the project work shall be based on the rubrics.
SEE: SEE will be conducted by the two examiners appointed by the Office of COE. The SEE marks awarded for the project work shall be based on the rubrics.

National Service Scheme / Physical Education / Yoga / Music / Fine Arts:

All students are required to register for any one of the following courses; National Service Scheme (NSS), Physical Education (PE) (Sports and Athletics), or Yoga (YOG), Music, Fine Arts —with the respective course coordinator during the first week of the third semester.

1) Department shall submit Continuous Internal Evaluation (CIE) marks for each semester based on the activities completed by students under the selected course.
2) Students may opt for different activities/options across semesters. For instance, a student participating in PE during 3rd semester may choose NSS in the 4th semester or Yoga.
3) Activities shall be conducted over two semesters (III & IV), and successful completion of the registered course / or courses along with the required CIE score is mandatory for the award of the degree.
4) Department and Institution must ensure that events are appropriately scheduled and reflected in the semester-wise calendar for NSS, PE, Yoga, Music and Arts activities.

These courses shall not be considered for the calculation of SGPA or CGPA and for vertical progression. However, completion of course(s) is compulsory for degree eligibility.

Bridge Courses on Mathematics for Lateral Entry Students:

This courses can be taught in the offline mode by the faculty of the mathematics department of the college as per the normal procedure to the students. All lateral entry students are required to register compulsorily for this course in the 3rd semester & 4th Semester and must appear for CIE. Passing in this course is mandatory for the award of the degree. Those who fail to secure the passing CIE marks, have to appear for the summer semester of the academic year or during subsequent odd semester. However, this course will not be considered for vertical progression, SGPA, and CGPA calculation.

Member Convenor - ISE **Chairperson - ISE** **Dean Academics** **Principal**

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SEMESTER: IV

Sl No	Course Category	Course Code	Course Title	Teaching Dept. (TD)	Paper Setting Dept. (PSD)	Mapping to SDG#	Teaching Learning Hours					Examination			CREDITS
							L	T	P	TW & SL	TOTAL NLH	Duration in hours	CIE: SEE Marks	Total Marks	
1	ASC	B25IS41	Discrete Mathematical Structures	IS	IS	4,9,11	42	0	0	48	90	03	50:50	100	3
2	IPCC	B25IS42	Database Management Systems	IS	IS	4,9,11	42	0	28	50	120	03	50:50	100	4
3	IPCC	B25CS43	Computer Networks	IS	CS	4,9,11	42	0	28	50	120	03	50:50	100	4
4	PCC	B25CS44	Analysis & Design of Algorithms	IS	CS	4,9,11	42	0	0	48	90	03	50:50	100	3
5	PCC	B25IS45	File Structures	IS	IS	4,9,11	42	0	0	48	90	03	50:50	100	3
6	PCCL	B25CSL46	Analysis & Design of Algorithms Laboratory	IS	CS	4,9,11	0	0	28	02	30	03	50:50	100	1
7	AEC	B25xxL47x	Ability Enhancement Course - II	IS	IS	4,9,11	0	0	28	02	30	03	50:50	100	1
8	BSC	B25CS48	Biology for Computer Engineers	IS	CS	3,4,8,13, 15	14	0	0	16	30	03	50:50	100	1
9	SDC	B25ISEP49	Environmental Science Project	IS	IS	4,7,9,11	0	0	0	30	30	03	50:50	100	1
10	NCMC	B25NSK410	National Service Scheme (NSS)	NSS Coordinator	EC	3,4,5,6, 13	0	0	0	28	28	-	100:0	100	-
		B25PEK410	Physical Education (PE) (Sports and Athletics)	PE Director	IS	3,5,8									
		B25YOK410	Yoga	Yoga Teacher	CI	3,4,12,16									
		B25MUK410	Music	Music Teacher	AD	3,4,11									

		B25FAK410	Fine arts	Arts Teacher	EE	4,5,8,11, 12									
11	NCMC	B25DIP411	Additional Mathematics II	MAT	MAT	4,8,9	28	0	0	20	48	-	100:0	100	-
12	NCMC	B25RRK412	Ready2Rise	T&P	T&P	4	14	0	0	14	28	-	100:0	100	-
												TOTAL	**750:450**	**1200**	**21**

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Ability Enhancement Course - II

Course Code	Course Title	Mapping to SDG#	Course Code	Course Title	Mapping to SDG#
B25ISL471	Web Design	4,9,11	B25ISL473	Technical Writing using LATEX	4,9
B25ISL472	Data Visualization using PYTHON	4,9,11			

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Skill Development Courses (SDC): These courses are intended to develop specific practical skills and competencies that improve students' employability, technical proficiency, and professional readiness to bridge the gap between academic and industry requirements. These courses emphasize hands-on training, application of theoretical knowledge, and development of discipline-relevant and transferable skills required in industry and society, and develop entrepreneurship and start-up skills.

Environmental Science Project

The Environmental Science Project is an applied learning component designed to develop students' awareness, understanding, and responsibility toward the environment. It provides an opportunity to study real-world environmental issues and apply scientific and engineering principles to design feasible and sustainable solutions.

The topics under environment include, but not limited to, climate change, biodiversity, air and water pollution, land use, excess use of natural resources, earthquakes, rise in the earth's temperature, power generation, soil erosion, environment issues related programmme, etc.

The project involves problem identification, field surveys, case studies, data collection, environmental audits, analysis, and proposal of remedial or preventive measures aimed at improving biodiversity, air quality, and thermal comfort, etc. Students can take up project individually or in a group not exceeding 4 students. Students can opt for Interdisciplinary Project based on their interest.

The evaluation shall be done as per the following;

CIE: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the project work, shall be based on the rubrics.
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7) Activities shall be conducted over two semesters (III & IV), and successful completion of the registered course / or courses along with the required CIE score is mandatory for the award of the degree.
8) Department and Institution must ensure that events are appropriately scheduled and reflected in the semester-wise calendar for NSS, PE, Yoga, Music and Arts activities.

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ASSESSMENT PATTERN

Table - 1: Typical distribution for Regular Theory (Non Integrated) Courses.

Covers courses whose credit breakup is 4:0:0:0 / 3:0:0:0 / 3:2:0:0

Assessment	Component	Marks	Reduced to	TOTAL Marks	Eligibility to appear for SEE
CIE	Internals - I	40	-	40 (Average of 3 test)	20
	Internals - II	40	-		
	Internals - III	40	-		
	AAT - I	10	-	Average out of 10	
	AAT - II	10	-		
SEE	Semester End Exam	100	50	50	
TOTAL MARKS				100	

Pattern of SEE Question Paper

Q#	Contents	Marks	Reduced to
PART A			50
1	Objective type of 1 mark / maximum of 2 marks questions. Total 3 marks from each module.	15	
PART B			
2 & 3	Module 1 – Question 2 or 3	17	
4 & 5	Module 2 – Question 4 or 5	17	
6 & 7	Module 3 – Question 6 or 7	17	
8 & 9	Module 4 – Question 8 or 9	17	
10 & 11	Module 5 – Question 10 or 11	17	
	MAXIMUM MARKS FOR SEE THEORY	100	

Table - 2: Typical distribution for Practical Courses / AEC Courses.

Covers courses whose credit breakup is 0:0:2:0 (not courses of Great Learning)

Assessment	Component	Marks	Reduced to	TOTAL Marks	Eligibility to appear for SEE
CIE	Lab - Continuous Internal Assessment (Weekly Performance)	20	-	50	20
	Internal Test	50	30		
SEE	Semester End Exam	100	50	50	
TOTAL MARKS				100	

Table – 3: Typical distribution for Integrated Courses.

Covers courses whose credit breakup is 3:0:2:0 / 2:0:2:0

Assessment	Component	Marks	Reduced to	TOTAL Marks	Eligibility to appear for SEE
CIE	Internals - I	40	Average of all 3 tests out of 40	Scaled down to 30	12
	Internals - II	40			
	Internals - III	40			
	Lab - Continuous Internal Assessment (Weekly Performance)	10	-	20	8
	Internal Test	50	10		
SEE	Semester End Exam	100	50	50	
TOTAL MARKS				100	

Pattern of SEE Question Paper

Q#	Contents	Marks	Reduced to
PART A			50
1	Objective type of 1 mark / maximum of 2 marks questions. Total 3 marks from each module.	15	
PART B			
2 & 3	Module 1 – Question 2 or 3	17	
4 & 5	Module 2 – Question 4 or 5	17	
6 & 7	Module 3 – Question 6 or 7	17	
8 & 9	Module 4 – Question 8 or 9	17	
10 & 11	Module 5 – Question 10 or 11	17	
MAXIMUM MARKS FOR SEE THEORY		100	

Table - 4: Typical distribution for ONE CREDIT COURSE.

Covers courses whose credit breakup is 1:0:0:0 (not courses of Great Learning)

Assessment	Component	Marks	TOTAL Marks	Eligibility to appear for SEE
CIE	Internals – I (MCQ based)	50	50 (Average)	20
	Internals – II (MCQ based)	50		
SEE	Semester End Exam (MCQ based)	50	50	
TOTAL MARKS			100	

Table - 5: Typical distribution for ONE CREDIT COURSE.

Covers courses whose credit breakup is 0:0:2:0 (courses of Great Learning)

Assessment	Component	Marks	Reduced to	TOTAL Marks	Eligibility to appear for SEE
CIE	Lab - Continuous Internal Assessment (Weekly Performance)	15	15	50	20
	Internal Lab Test	50	15		
	Proctored (Certification) Exam	100	20		
SEE	Semester End Exam	100	50	50	
TOTAL MARKS				100	

Table – 6: Typical distribution for NON CREDIT MANDATORY COURSES.

Covers courses whose credit breakup is 0:0:0:0 (basically PP courses)

Assessment	Component	Marks	TOTAL Marks	Eligibility to PASS
CIE	Internals – I (MCQ based)	50	100	40
	Internals – II (MCQ based)	50		
TOTAL MARKS			100	

Table – 7: Typical distribution for PROJECT related SDC (having 1 / 2 credits)

Ex: Community Project / Societal Project / Environmental Science Project

Assessment	Component	Marks	TOTAL Marks
CIE	Review - I	20	50
	Review – II	30	
SEE	Semester End Exam	50	50
TOTAL MARKS			100

Table – 8: Typical distribution for MINI PROJECT (having 3 credits) / MAJOR / CAPSTONE PROJECT PHASE – I (having 3 credits)

Assessment	Component	Marks	TOTAL Marks
CIE	Review - I	20	100
	Review – II	30	
	Review – III	50	
TOTAL MARKS			100

Table – 9 Typical distributions for MAJOR / CAPSTONE PROJECT PHASE – II (having 6 / 7 credits) and INTERNSHIP (having 9 / 10 credits)

Assessment	Component	Marks	TOTAL Marks
CIE	Review – I	20	100
	Review – II	30	
	Review – III	50	
SEE	Semester End Exam	100	100
TOTAL MARKS			200

Table – 10: Typical distribution for NON CREDIT MANDATORY COURSES.

Covers courses whose credit breakup in terms of hours is 14:0:0:14 (basically PP courses)

Example: Mathematics Course for Lateral Entry students (CIE:100; SEE:0)

Assessment	Component	Marks	TOTAL Marks	Eligibility to PASS
CIE	Internals - I	40	100	40
	Internals - II	40		
	AAT-I	10		
	AAT-II	10		
TOTAL MARKS			100	

Table – 11: Typical distribution for Innovation and Design Thinking Lab.

CIE:50 SEE:50 – Total: 100 Marks

CIE Parameters: 50 Marks

Sl. No.	CIE Component / Week	Marks
1	Orientation Activities & Communication Skills	05
2	Empathy & Field Exploration Documentation	10
3	Problem Definition and Framing	10
4	Ideation & Mind Mapping	10
5	Prototype Development & Iteration	10
6	Teamwork, Journal, and Engagement	05
	Total CIE marks	**50**

SEE Parameters: 100 Marks

Sl. No.	CIE Component / Week	Marks
1	Prototype / Solution Demonstration	30
2	Final Presentation / Social Pitch	20
3	Business Model or Sustainability Plan	10
4	Viva Voce	20
5	Documentation Report / Portfolio	20
	Total SEE marks	**100**

Table – 12: Typical distribution for Interdisciplinary Project Based Learning.

CIE:50 SEE:50 – Total: 100 Marks

CIE Parameters: 50 Marks

Sl. No.	CIE Component / Week	Marks
1	Problem Identification & Literature Survey	05
2	Concept Development & Design	10
3	Prototype Development	15
4	Testing & Validation	05
5	Documentation & Presentation	15
	Total CIE marks	**50**

SEE Parameters: 100 Marks

Sl. No.	CIE Component / Week	Marks
1	Final Presentation & Demonstration	30
2	Prototype Quality & Functionality	30
3	Documentation Report	20
4	Social Impact & Sustainability	05
5	Innovation & Originality	10
6	Viva – voce	05
	Total SEE marks	**100**

PROBABILITY AND STATISTICS

Course Code	B25MAT31A	**Total contact hours / week**	05	**CIE MARKS**	50
Course Category	ASC	**Total SAAE Hours / semester**	50	**SEE MARKS**	50
L:T:P:S	3:2:0:0	**Total Notional Learning Hours**	120	**TOTAL MARKS**	100
Total credits	04			**Exam Duration**	03 Hrs
Mapping to SDG #	4, 8,9,11 and 17	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 8:** Decent work and Economic Growth **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities **SDG 17:** Partnership for the goals		

COURSE LEARNING OBJECTIVES
1. To introduce the concept of random variables, probability distributions, discrete and continuous distributions with practical application in Computer Science Engineering and real- life situations. 2. To provide the principles of statistical inferences and the basics of hypothesis testing with emphasis on some commonly encountered hypotheses. 3. To determine whether an input has a statistically significant effect on the system's response through ANOVA testing.

PRE-REQUISITES
1. Fundamentals of Statistics and Probability Theory

MODULE #	TOPICS	Hours
1	**Probability Distributions** Review of basic probability theory. Random variables (discrete and continuous), probability mass and density functions. Mathematical expectation, mean and variance. Binomial, Poisson, Exponential and normal distributions- problems (derivations for mean and standard deviation for Binomial and Poisson distributions only)-Illustrative examples. **Textbook 1:Chapter 5.1 - 5.4, 7.1, 7.3, 8.2, 8.6**	9L+5T
	PRACTICAL: 1) Illustrate working of Binomial distribution, Poisson distribution, Exponential distribution and Normal distribution using Python.	
	RESEARCH: 1) Apply probability distributions to predict the possibilities of an event for a real-life problem.	
	Joint Probability Distributions: Joint Probability distribution for two discrete random variables, expectation, covariance and correlation. **Markov Chains:** Introduction to Stochastic Process, Probability Vectors, Stochastic matrices, Regular stochastic matrices, Markov chain, Higher transition probabilities, Stationary distribution of Regular Markov chain and absorbing states.	9L+5T

2	**Textbook 1:Chapter 5.5**	
	PRACTICAL: 1) Write a mathematical code to i) evaluate probabilities of discrete joint probability distributions, ii) obtain fixed probability vector of a transition probability matrix using Python.	
	RESEARCH: 1) Explore the areas where stochastic processes take place and explain how they give rise to Generative AI and other machine learning algorithms.	
3	**Introduction to Data Analytics:** Review of measures of central tendency, Quartiles, Range, Interquartile range, Moments, Skewness, measures of skewness. Kurtosis and measures of Kurtosis, Correlation coefficient, Regression coefficient, linear regression. **Textbook 1:Chapter 3.5 - 3.9, 3.13, 3.14, 10.1 -10.7**	8L+6T
	PRACTICAL: 1) Write a mathematical code to i) evaluate moments, skewness ii) correlation and regression coefficients using Python.	
	RESEARCH: 1) Use Linear Regression model for a case study and test the adequacy of the model.	
4	**Statistical Inference:** Introduction, sampling distribution, standard error, testing of hypothesis, levels of significance, test of significances, confidence limits, simple sampling of attributes, test of significance for large samples, comparison of large samples. Sampling variables, central limit theorem and confidence limits for unknown means. Test of Significance for means of two small samples, students 't' distribution and Chi-square distribution as a test of goodness of fit. **Textbook 1:Chapter 12.1 - 12.9, 13.7.2, 14.1, 14.2**	8L+6T
	PRACTICAL 1) Apply the Z-test, t-test, Chi-Square test appropriately to test the hypothesis at a given level of significance using pen, paper and Python.	
	RESEARCH: 1) Make a literature survey on different types of sampling techniques for data collection which helps in making best decisions.	
5	**Analysis Of Variance:** Basic principles of ANOVA, One way analysis of variance, Two way analysis of variance, Latin-square Design - problems only **Textbook 2: Chapter 11**	8L+6T
	PRACTICAL: 1) Write a mathematical code to carry out one-way analysis of variance and two-way analysis of variance techniques using Python.	
	RESEARCH: 1) Construct randomized block designs and Latin Square Designs for a real life situation and carry out ANOVA.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Fundamentals of Mathematical Statistics	S C Gupta and V K Kapoor	10th	S Chand and Company	2002
2	Research Methodology Methods and Techniques	Kothari C R	2nd	New age International Publisher	2009

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Advanced Engineering Mathematics	E. Kreyszig	10th	John Wiley & Sons	2016
2	Probability, Queuing Theory & Reliability Engineering	G. Haribaskaran	2nd	Laxmi Publication	2006
3	Mathematical Statistics with Applications	Irwin Miller & Marylees Miller	8th	Pearson. Dorling Kindersley Pvt. Ltd.	2014
4	Practical Statistics for Data Scientists	Peter Bruce, Andrew Bruce & Peter Gedeck	2nd	O'Reilly Media, Inc.	2020
5	A Text book of Engineering Mathematics	N.P Bali and Manish Goyal	10th	Laxmi Publications	2022
6	Higher Engineering Mathematics	B.S. Grewal	44th	Khanna Publishers	2021

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**NPTEL**	http://nptel.ac.in/courses.php?disciplineID=111
2	**MOOCs**	http://www.class-central.com/subject/math
3	**E-VIDEO**	http://academicearth.org/
4	**MOOCs**	http://www.class-central.com/subject/math(MOOCs)
5	**Kaggle**	https://www.kaggle.com/discussions/getting-started/111585
6	**Kaggle**	https://www.kaggle.com/code/hamelg/python-for-data-26-anova
7	**Karnataka Agriculture Department**	https://raitamitra.karnataka.gov.in/126/Agriculture%20Statistics/en

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply Binomial, Poisson, Exponential and Normal distributions to find the possibilities of uncertain events using pen, paper and Python.
CO2	Explain the concept of Joint Probability Distributions, discrete-time Markov chain and n-step transition probabilities for problems arising in Engineering using pen, paper and Python.
CO3	Apply the various statistical measures and analyze huge volumes of data using pen, paper and Python.
CO4	Apply the Z-test, t-test, and Chi-Square test appropriately to test the hypothesis at a given level of significance.
CO5	Apply the ANOVA test for problems arising in various fields of science, Engineering and technology using pen, paper and Python.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	2	2	-	-	2	2	-	2	2	2	-
CO2	3	3	-	2	2	-	-	2	2	-	2	2	2	-
CO3	3	3	-	2	2	-	-	2	2	-	2	2	2	-
CO4	3	3	-	2	2	-	-	2	2	-	2	2	2	-
CO5	3	3	-	2	2	-	-	2	2	-	2	2	2	-
TOTAL	15	15		10	10			10	10		10	10	10	-
AVG	3	3		2	2			2	2		2	2	2	-

Teaching-Learning Process
Pedagogy (General Instructions):
These are sample Strategies; which teachers can use to accelerate the attainment of the various course outcomes.

1. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the delivered lessons shall develop students' theoretical and applied mathematical skills.
2. State the need for Mathematics with Engineering Studies and provide real-life examples.
3. Support and guide the students for self–study.
4. You will also be responsible for assigning homework, grading assignments quiz, and documenting student's progress.
5. Encourage the students to group learning to improve their creative and analytical skills.

OBJECT ORIENTED PROGRAMMING USING JAVA

Course Code	B25IS32	**Total contact hours / week**	03	**CIE MARKS**	50
Course Category	IPCC	**Total SAAE Hours / semester**	50	**SEE MARKS**	50
L: T:P:S	3:0:2:0	**Total Notional Learning Hours**	120	**TOTAL MARKS**	100
Total credits	04			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 8 and 9	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 8:** Decent Work and Economic Growth **SDG 9:** Industry, Innovation, and Infrastructure		

COURSE LEARNING OBJECTIVES
1. To explain the fundamentals of Java programming including data types, variables, arrays, operators, and control statements. 2. To describe classes, objects, methods, constructors, and object-oriented programming features in Java. 3. To discuss inheritance, polymorphism, abstract classes, and interfaces for developing reusable software components. 4. To illustrate packages and exception-handling mechanisms for developing modular and reliable Java applications. 5. To summarize multithreading concepts and concurrent programming techniques in Java.

PREREQUISITES
1. Basics of C Programming.

MODULE #	TOPICS	Hours
1	**An Overview of Java:** Object-Oriented Programming, Using Blocks of Code, Lexical Issues **Data Types, Variables, and Arrays:** The Primitive Types, Variables, Type Conversion and Casting, Automatic Type Promotion in Expressions, Arrays, Introducing Type Inference with Local Variables. **Operators:** Arithmetic Operators, Relational Operators, Boolean Logical Operators, The Assignment Operator, The? Operator, Operator Precedence, Using Parentheses. **Control Statements:** Java's Selection Statements, Iteration Statements, Jump Statements. **Textbook 1: Chapter 2, 3, 4, 5**	9
	PRACTICAL: 1) Demonstrate primitive data types in Java 2) Program for type conversion and type casting 3) Program to demonstrate arithmetic operators 4) Program using assignment and ternary operators 5) Program to show operator precedence	

	RESEARCH: 1) Features of Java and platform independence 2) JVM architecture and working 3) Importance of abstraction in software development	
2	**Introducing Classes:** Class Fundamentals**,** Declaring Objects**,** Assigning Object Reference Variables**,** Introducing Methods**,** Constructors**,** This Keyword**,** Garbage Collection. **Methods and Classes:** Overloading Methods, Objects as Parameters, Argument Passing, Returning Objects, Recursion, Access Control, understanding static, introducing final, Introducing Nested and Inner Classes. **Textbook 1: Chapter 6, 7**	8
	PRACTICAL: 1) Create and call methods 2) Demonstrate constructor overloading 3) Implement method overloading 4) Write recursive program	
	RESEARCH: 1) Importance of classes and objects in OOP 2) Compile-time polymorphism using method overloading 3) Memory optimization using static members	
3	**Inheritance:** Inheritance Basics, using super, creating a Multilevel Hierarchy, When Constructors Are Executed, Method Overriding, Dynamic Method Dispatch, Using Abstract Classes, Using final with Inheritance, Local Variable Type Inference and Inheritance, The Object Class. **Interfaces:** Interfaces, Default Interface Methods, Use static Methods in an Interface, Private Interface Methods. **Textbook 1:Chapter 8, 9**	8
	PRACTICAL: 1) Create programs demonstrating inheritance 2) Use the super keyword 3) Demonstrate dynamic method dispatch 4) Create and implement interfaces 5) Use static methods in interfaces	
	RESEARCH: 1) Importance of inheritance in software reuse 2) Constructor chaining in inheritance	
4	**Packages:** Packages, Packages and Member Access, Importing Packages. **Exceptions:** Exception-Handling Fundamentals**,** Exception Types**,** Uncaught Exceptions**,** using try and catch, Multiple catch Clauses, Nested try Statements, throw, throws, finally, Java's Built-in Exceptions, Creating Your Own Exception Subclasses, Chained Exceptions. **Textbook 1: Chapter 9, 10**	9
	PRACTICAL: 1) Demonstrate package hierarchy 2) Access package members using access modifiers 3) Handle multiple exceptions using multiple catch blocks 4) Demonstrate nested try statements 5) Handle built-in exceptions 6) Demonstrate chained exceptions	
	RESEARCH: 1) Importance of packages in Java programming 2) Fundamentals of exception handling in Java 3) Chained exceptions and debugging	

5	**Multithreaded Programming:** The Java Thread Model, The Main Thread, creating a Thread, Creating Multiple Threads, using is Alive () and join (), Thread Priorities, Synchronization, Interthread Communication, Suspending, Resuming, and Stopping Threads, Obtaining a Thread's State. **Textbook 1: Chapter 11**	8
	PRACTICAL: 1) Demonstrate the Java thread model 2) Create threads using Thread class 3) Implement thread synchronization 4) Demonstrate suspending, resuming, and stopping threads	
	RESEARCH: 1) Importance of multithreading in Java 2) Applications of multithreading in real-world systems	

PRACTICAL COMPONENT OF IPCC

Sl. No.	LIST OF PROGRAMS
1	Write a program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula.
2	Create a Java class called Student with the following details as variables within it. USN Name Branch Phone Write a Java program to create n Student objects and print the USN, Name, Branch, and Phone of these objects with suitable headings.
3	Write a program to check whether the given number is prime or not.
4	Design a super class called Staff with details as Staff ID, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (skills), and Contract (period). Write a Java program to read and display at least 3 staff objects of all three categories.
5	Write a program demonstrating Method overloading and Constructor overloading.
6	Develop a java application to implement currency converter (Dollar to INR, EURO to INR, Yen to INR and vice versa), distance converter (meter to KM, miles to KM and vice versa), time converter (hours to minutes, seconds and vice versa) using packages.
7	Write a program to generate the resume. Create 2 Java classes Teacher (data: personal information, qualification, experience, achievements) and Student (data: personal information, result, discipline) which implements the java interface Resume with the method biodata ().
8	Write a program that implements a multi-thread application that has three threads. First thread generates a random integer for every 1 second; second thread computes the square of the number and prints; third thread will print the value of cube of the number.
9	Write a program to perform string operations using Array List. Write functions for the following a. Append - add at end b. Insert – add at particular index c. Search d. List all string starts with given letter.
10	Write a program to read two integers a and b. Compute a/b and print, when b is not zero. Raise an exception when b is equal to zero.

PRESCRIBED TEXTBOOKS

Textbook #	Book Title	Authors	Edition	Publisher	Year
1	Java: The Complete Reference	Herbert Schildt	13th	McGraw-Hill	2024

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Programming with Java	E Balagurusamy	6th	McGraw Hill	2022
2	Thinking in Java	Bruce Eckel	4th	Prentice Hall	2006

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**NPTEL**	https://nptel.ac.in/courses/106105191
2	**YouTube**	https://www.youtube.com/watch?v=xk4_1vDrzzo https://www.youtube.com/watch?v=Ej_02ICOIgs https://www.youtube.com/watch?v=Zs342ePFvRI https://www.youtube.com/watch?v=0ik6X4DJKCc https://www.youtube.com/watch?v=W-N2ltgU-X4
3	**GeeksforGeeks**	https://www.geeksforgeeks.org/java/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Demonstrate proficiency in writing simple programs involving branching and looping structures.
CO2	Design a class involving data members and methods for the given scenario.
CO3	Apply the concepts of inheritance and interfaces in solving real world problems.
CO4	Use the concept of packages and exception handling in solving complex problem.
CO5	Apply concepts of multithreading in program development.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	-	2	-	-	-	1	-	-	1	2	-
CO2	3	2	2	-	2	-	-	-	2	-	-	1	2	-
CO3	3	2	3	-	2	-	-	-	2	-	-	1	2	-
CO4	3	2	2	1	2	-	-	-	1	-	-	1	2	-
CO5	3	2	2	1	3	-	-	-	2	-	-	1	2	-
TOTAL	15	10	10	2	11	-	-	-	8	-	-	5	10	-
AVG	3	2	2	1	2.2	-	-	-	1.6	-	-	1	2	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Chalk and Talk.
2) ICT based Teaching.
3) Demonstration based Teaching.

OPERATING SYSTEMS

Course Code	B25CI33	**Total contact hours / week**	04	**CIE MARKS**	50
Course Category	IPCC	**Total SAAE Hours / semester**	50	**SEE MARKS**	50
L:T:P:S	3:0:2:0	**Total Notional Learning Hours**	120	**TOTAL MARKS**	100
Total credits	4			**Exam Duration**	03 Hrs
Mapping to SDG#	4,9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industrial Innovation and Infrastructure **SDG 11:** Sustainable cities and communities		

COURSE LEARNING OBJECTIVES
1. To Demonstrate the need for OS and different types of OS. 2. To Discuss suitable techniques for management of different resources. 3. To Demonstrate different APIs/Commands related to processor. 4. To Demonstrate different APIs/Commands related to memory. 5. To Demonstrate different APIs/Commands related to storage and file system management.

PREREQUISITES
1. Knowledge of Computer system with CPU, Memory and I/O Devices. 2. Knowledge of RAM, ROM and Hard Disk of a computer system. 3. Basic fundamentals of Hardware and Software.

MODULE #	**TOPICS**	**Hours**
1	**INTRODUCTION TO OPERATING SYSTEMS AND SYSTEM STRUCTURES:** What operating systems do; Computer System organization; Computer System architecture; Operating System structure; Operating System operations; Process management; Memory management; Storage management; Protection and Security; Distributed system; Special-purpose systems; Computing environments. **OPERATING SYSTEM SERVICES:** User - Operating System interface; System calls; Types of system calls; Virtual machines; System boot. **Textbook 1: Chapter 1(1.1-1.12), Chapter 2(2.2,2.3,2.4,2.8,2.11)**	8
	PRACTICAL: 1) Implement fork (), exec (), wait (), and sleep () system calls. 2) Implement soft link and hard link for files and compare their behaviour. 3) Arrange processes/files in ascending and descending order of size using UNIX commands and shell scripts.	
	RESEARCH: 1) Designing an Energy-Efficient Operating System Framework for AI- based Edge Devices in Smart Cities**.**	

2	**PROCESS MANAGEMENT:** **Process concept:** Process scheduling; Operations on processes; Inter-process communication. **MULTI-THREADED PROGRAMMING:** Overview; Multithreading models; Threading issues. **PROCESS SCHEDULING:** Basic concepts; Scheduling criteria; Scheduling algorithms; Thread scheduling; Multiple-processor scheduling. **Textbook 1: Chapter 3 (3.1 - 3.4), Chapter 4 (4.1 - 4.4), Chapter 5 (5.1 - 5.5)**	8
	PRACTICAL: 1) Simulate CPU scheduling algorithm using FCFS. 2) Simulate CPU scheduling algorithms using Round Robin, Priority, and SJF. 3) Implement protection of a process from Read, Write, and Execute. permissions. Implement process creation and process kill operations.	
	RESEARCH: 1) AI-Driven Adaptive CPU Scheduling for Real-Time Multitasking Systems.	
3	**PROCESS SYNCHRONIZATION:** Synchronization; The Critical section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical synchronization problems. **DEADLOCKS:** System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock. **Textbook 1:Chapter 6 (6.1 - 6.6), Chapter (7.1 - 7.7)**	9
	PRACTICAL: 1) Implement the Producer–Consumer problem using a semaphore. 2) Implement the Reader–Writer problem. 3) Implement Banker's algorithm for deadlock avoidance.	
	RESEARCH: **1)** Machine Learning-Based Prediction and Prevention of Deadlocks in Concurrent Operating Systems.	
4	**MEMORY MANAGEMENT:** Memory management strategies Background; Swapping; Contiguous memory allocation; Paging; Structure of page table; Segmentation. **VIRTUAL MEMORY MANAGEMENT:** Background; Demand paging; Copy-on-write; Page replacement; Allocation of frames; Thrashing. **Textbook 1:Chapter 8 (8.1 - 8.6), Chapter 9 (9.1 - 9.6)**	8
	PRACTICAL: 1) Implement memory management techniques using First Fit, Best Fit, and Worst Fit. 2) Simulate page replacement algorithms FIFO and LRU.	
	RESEARCH: 1) Predictive Page Replacement Using Machine Learning for Efficient Memory Management.	
5	**FILE SYSTEM, IMPLEMENTATION OF FILE SYSTEM:** File system: File concept; Access methods; Directory and Disk structure; **IMPLEMENTATION OF FILE SYSTEM:** File system structure; File system implementation; Directory implementation; Allocation methods. **SECONDARY STORAGE STRUCTURE, PROTECTION:** Mass storage structures; Disk structure; Disk attachment; Disk scheduling; Disk management. **PROTECTION:** Goals of protection; Principles of protection; Domain of protection; Access matrix.	9

	Textbook 1: Chapter 10 (10.1 - 10.3), Chapter 11 (11.1 - 11.4), Chapter 12(12.1 - 12.5) ,Chapter 14 (14.1 - 14.4)	
	1) **PRACTICAL:** Simulate file organization and file allocation techniques: Single-level directory and Linked file allocation. 2) Implement FCFS and SCAN disk scheduling algorithms.	
	RESEARCH: 1) Intelligent File Allocation and Disk Scheduling for Large-Scale Data Systems Using AI.	

PRACTICAL COMPONENT OF IPCC

Sl. No.	LIST OF PROGRAMS
1	Develop a C program to implement the Process system calls (fork(), exec(), wait(), create process, terminate process).
2	Simulate the following CPU scheduling algorithms to find turnaround time and waiting time: a) FCFS b) SJF c) Round Robin d) Priority
3	Develop a C program to simulate producer-consumer problem using semaphores.
4	Develop a C program which demonstrates interprocess communication between a reader process and a writer process. Use mkfifo (), open, read, write and close APIs in your program.
5	Develop a C program to simulate Bankers Algorithm for Deadlock Avoidance.
6	Develop a C program to simulate the following contiguous memory allocation Techniques: a) Worst fit b) Best fit c) First fit
7	Develop a C program to simulate page replacement algorithms: a) FIFO b) LRU
8	Simulate following File Organization Techniques: a) Single level directory b) Two level directory
9	Develop a C program to simulate the Linked file allocation strategies.
10	Develop a C program to simulate SCAN disk scheduling algorithm

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Operating System Concepts	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne	8th	Wiley- India	2015

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Understanding Operating System.	Ann McHoes, Ida M. Fylnn	8th	Cengage Learning	2018
2	Operating Systems: A Concept Based Approach.	D.M Dhamdhere	3rd	McGraw-Hill	2013
3	An Introduction to Operating Systems: Concepts and Practice.	P.C.P. Bhatt	4th	PHI(EEE)	2014
4	Operating Systems: Internals and Design Principles.	William Stallings,	6th	Pearson.	-

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**You Tube - Dr. Mike Murphy Lecture Series**	https://youtu.be/mXw9ruZaxzQ.
2	**You Tube-Introduction to OS**	https://youtu.be/vBURT97EkA.
3	**You Tube -Gate Smashers**	https://www.youtube.com/watch?v=783KAB-tuE4&list=PLIemF3uozcAKTgsCIj82voMK3TMR0YE_f.
4	**Operating Systems Complete Free Online Course**	https://www.youtube.com/watch?v=3-lTLMMeeXY&list=PL3pGy4HtqwD0n7bQfHjPnsWzkeRn6mkO.

COURSE OUTCOMES:

At the end of the course, student will be able to:

CO #	Course Outcome Statement
CO1	Explain the structure and functionality of operating system.
CO2	Apply appropriate CPU scheduling algorithms for the given problem.
CO3	Analyse the various techniques for process synchronization and deadlock handling.
CO4	Apply the various techniques for memory management.
CO5	Explain file and secondary storage management strategies
CO6	Describe the need for information protection mechanisms.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	-	-	3	3	-	-	2	-	-	-
CO2	3	3	2	-	-	-	3	3	-	-	2	3	2	-
CO3	3	3	2	-	-	-	3	3	-	-	2	3	2	-
CO4	3	3	2	-	-	-	3	3	-	-	2	3	2	-
CO5	3	3	2	-	-	-	3	3	-	-	2	-	-	-
CO6	3	3	2	-	-	-	3	3	-	-	2	-	-	
TOTAL	18	18	12	-	-	-	18	18	-	-	12	9	6	-
AVG	3	3	2	-	-	-	3	3	-	-	2	3	2	-

Teaching-Learning Process (General Instructions):

Teachers can use the following strategies to accelerate the attainment of the various course outcomes.

1. Lecturer methods (L) need not to be only traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
5. Role play for process scheduling.
6. Demonstrate the installation of any one Linux OS on VMware/Virtual Box**.**

DATA STRUCTURES

Course Code	B25CS34	**Total contact hours / week**	04	**CIE MARKS**	50
Course Category	PCC	**Total SAAE Hours / semester**	48	**SEE MARKS**	50
L:T:P:S	3:0:0:0	**Total Notional Learning Hours**	90	**TOTAL MARKS**	100
Total credits	03			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities:		

COURSE LEARNING OBJECTIVES
1. To explain fundamentals of data structures and their applications. 2. To understanding and apply different sorting techniques. 3. To illustrate representation of Different data structures such as Stack, Queues, Linked Lists, Trees and Graphs. 4. To Design and Develop Solutions to problems using Linear and Nonlinear Data Structures.

PREREQUISITES
1. Basics of C programming

MODULE #	TOPICS	Hours
1	**INTRODUCTION TO DATA STRUCTURES:** Data Structures, Classification of Data Structures (Primitive & Non-Primitive), Operations on Data structure **Sparse Matrix**-Abstract data type, sparse matrix representation, transposing matrix. **SORTING TECHNIQUES-** Merge sort, Quick sort, Radix sort, insertion sort. **HASHING**: Hash functions, collision-chaining, open addressing-linear probing, quadratic probing, double hashing. **Textbook 1: Chapter 2(**2.1 - 2.3), **Chapter 14(**14.8- 14.12), **Chapter 15(**15.3, 15.4, 15.5.1) **Textbook 2**: **Chapter 2** (2.5.1,2.5.2,2.5.3)	8
	PRACTICAL: 1) Representation of Sparse matrix 2) Programs related to various sorting algorithms 3) Implement hash table 4) Implement hash table with double hashing 5) Implement hash tables chaining with linked list / doubly linked list / binary tree 6) Implement hash tables chaining with linear probing / quadratic probing	
	RESEARCH: 1) Recent Developments in Data Structures: Cache – Oblivious Data 2) Structures, Sketch Data Structures, Geometric Data Structures, Succinct	

	3) Data Structures, Self-Organizing Data Structures 4) Comparative Analysis of Sparse Matrix Algorithms for Information Retrieval 5) Performance Evaluation of Sorting Techniques 6) Hashing in Cryptography	
2	**STACKS:** Introduction, array Representation of stack, Basic operation of stack, Applications - Conversion of Expressions, Evaluation of expression. **QUEUES:** Basic operation of queue, Types of queues- Circular Queues, priority and Double ended queue. **Textbook 1**: **Chapter 7(** 7.1-7.3, 7.7-7.7.3), **Chapter 8(** 8.2, 8.4- 8.4.1, 8.4.2, 8.4.3)	9
	PRACTICAL: 1) Implement Stack using Arrays 2) Reverse stack using recursion 3) Stack operations using MACROS / Dynamic Memory Allocation 4) Check expression is correctly parenthesized 5) Tower of Hanoi 6) Conversion of Expressions / Evaluation of Expressions 7) Implement Queues using Arrays 8) Implement types of Queues 9) Queue operations using Dynamic Memory Allocation 10) Josephus problem	
	RESEARCH: 1) Stack data structure implemented in DNA 2) Queuing Theory in Cloud Computing	
3	**LINKED LISTS:** Dynamic Memory Allocation, Linked Lists- Introduction, Singly Linked Lists (SLL), Circular Linked Lists, Doubly Linked Lists(DLL), Circular DLL, Application – Linked Stacks and Queues, Polynomials- Polynomial Representation, Adding Polynomials. **Textbook 1: Chapter 6(**6.1 - 6.5) **Textbook 2: Chapter 1(**1.2-1.2.2), **Chapter 4** (4.3, 4.4-4.4.1, 4.4.2)	8
	PRACTICAL: 1) Perform primitive operations on single linked list 2) Perform primitive operations on doubly linked list 3) Stack implementation on linked list 4) Queue implementation on linked list 5) Reverse a linked list 6) Addition of two long integers 7) Addition of two polynomials	
	RESEARCH: 1) Object oriented paradigm approach to multi linked lists	
4	**TREES:** Introduction, Types of Trees - Binary Trees, Binary Search Trees, Expression Trees, creating a Binary Tree from a General Tree, traversing a Binary Tree, Binary Search Trees, Threaded Binary Trees, Applications of Trees. **Textbook 1: Chapter 9(** 9.1, 9.2-9.2.3, 9.2.4, 9.2.5, 9.3, 9.4, 9.6), **Chapter 10(** 10.1, 10.3)	9
	PRACTICAL: 1) Various Traversals 2) Threaded Binary Tree 3) Binary Search Tree (BST) 4) Primitive operations on BST 5) Sum of all nodes of a BST	

	6) Traversal without recursion	
	RESEARCH: 1) Hierarchical Clustering - Building Taxonomy Trees in Production AI	
5	**GRAPHS:** Introduction, Graph Terminology, Directed Graphs, Bi-connected Components, Representation of Graphs - Adjacency Matrix Representation and Adjacency List Representation, Graph Traversal Algorithms - Breadth-First Search Algorithm, Depth-first Search Algorithm, Application - topological sort. **Textbook 1: Chapter 13(**13.1- 13.4, 13.5-13.5.1, 13.5.2, 13.6, 13.7)	8
	PRACTICAL: 1) Breadth First Search 2) Depth First Search 3) Topological Sorting	
	RESEARCH: 1) Graph Theory Applications in Social Networks	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Data Structures using C	Reema Thareja	3rd	Oxford press,	2023
2	Fundamentals of Data Structures in C	Ellis Horowitz and Sartaj Sahni, ,	2nd	Universities Press	2018

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Data Structures: A Pseudo-code approach with C .	Gilberg and Forouzan	2nd	Cengage Learning	2014
2	Programming in ANSI C.	E. Bala Guruswamy,	9th	McGraw Hill.	2024
3	Data Structures using C.	A M Tenenbaum,	7th	PHI,	2018
4	Data Structures and Program Design in C.	Robert Kruse	2nd	PHI	2006
5	Data Structures using C.	Seymour	13th	McGraw Hill Special Indian	2017

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**VTU E-learning**	http://elearning.vtu.ac.in/econtent/courses/video/CSE/06CS35.html
2	**Swayam NPTEL**	https://nptel.ac.in/courses/106/105/106105171/ http://www.nptelvideos.in/2012/11/data-structures-and-algorithms.html
3	**Virtual labs**	https://ds2-iiith.vlabs.ac.in/exp/selection-sort/index.html
4	**Geeks for geeks**	https://www.geeksforgeeks.org/data-structures-and-algorithms-online-courses-free-and-paid/
5	**Simply learn**	https://www.simplilearn.com/free-data-structures-algorithms-course-skillup

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply array concepts to sort the elements, sparse matrix, hashing and hashing-based sorting techniques.
CO2	Develop programs on operations of stack and its applications, queue operations.
CO3	Apply the concepts of singly linked lists, doubly linked lists, circular linked lists into different applications.
CO4	Illustrate the construction of binary trees, binary search tress, and its traversal techniques.
CO5	Develop solutions using graphs to model the real-world problem.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	-	-	-	-	1	1	2	3	3	-
CO2	3	3	2	-	-	-	-	-	1	1	2	3	3	-
CO3	3	3	2	-	-	-	-	-	1	1	2	3	3	-
CO4	3	3	2	-	-	-	-	-	1	1	2	3	3	-
CO5	3	3	2	-	-	-	-	-	1	1	2	3	3	-
TOTAL	15	15	10	-	-	-	-	-	5	5	10	15	15	-
AVG	3	3	2	-	-	-	-	-	1	1	2	3	3	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Chalk and Talk with Black Board.
2) ICT based Teaching.
3) Demonstration based Teaching.

DIGITAL DESIGN AND COMPUTER ORGANIZATION

Course Code	B25IS35	**Total contact hours / week**	03	**CIE MARKS**	50
Course Category	PCC	**Total SAAE Hours / semester**	48	**SEE MARKS**	50
L:T:P:S	3:0:0:0	**Total Notional Learning Hours**	90	**TOTAL MARKS**	100
Total credits	03			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. To demonstrate the functionalities of binary logic system. 2. To explain the working of combinational and sequential logic system. 3. To realize the basic structure of computer system. 4. To illustrate the working of I/O operations and processing unit.

PREREQUISITES
1. Basic Knowledge of Number Systems. 2. Fundamentals of Boolean Algebra, Arithmetic operations, Logic Gates and basic logic circuit design. 3. Familiarity with computer components and their basic functions (CPU, memory, input/output).

MODULE #	TOPICS	Hours
1	**Introduction to Digital Design** Binary Logic, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Digital Logic Gates, Introduction, The Map Method, Four-Variable Map, Don't-Care Conditions, NAND and NOR Implementation, Other two level implementations. **Text book 1: Chapter 1(1.9), Chapter 2 (2.4, 2.5, 2.8), Chapter 3 (3.1, 3.2, 3.3, 3.5, 3.6, 3.7)**	8
	PRACTICAL: 1) Design of Simple Boolean logic using basic gates 2) Design of simple Boolean logic using Universal gates 3) Reduction of Boolean expressions and implement using basic gates .	
	RESEARCH: 1) Digital CMOS VLSI design. 2) Digital CMOS design using HDLs in VLSI	

2	**Combinational Logic:** Introduction, Combinational Circuits, Design Procedure, Binary Adder-Subtractor, Decoders, Encoders, Multiplexers. **Sequential Logic:** Introduction, Sequential Circuits, Storage Elements: Latches, Flip-Flops. **Text book 1: Chapter 4 (4.1, 4.2, 4.4, 4.5, 4.9, 4.10, 4.11), Chapter 5 (5.1, 5.2, 5.3, 5.4)**	9
	PRACTICAL: 1) Logical Design of complex Boolean circuits 2) Design and testing of Adder and Subtractor circuits with basic gates 3) Design and testing of Encoder and Decoder circuits using basic gates 4) Design and testing of Multiplexer and Demultiplexer circuits using basic gates 5) Design of AND-OR-Invert and OR-AND-Invert circuits	
	RESEARCH: 1) VLSI design of combinational circuits using CMOS logic 2) VLSI design of sequential circuits using CMOS logic 3) FANIN and FANOUT control in VLSI design for driver logic	
3	**Basic Structure of Computers:** Functional Units, Basic Operational Concepts, Bus structure, Performance – Processor Clock, Basic Performance Equation, Clock Rate, Performance Measurement. **Machine Instructions and Programs:** Memory Location and Addresses, Memory Operations, Instruction and Instruction sequencing, Addressing Modes. **Text book 2: Chapter 1(1.2, 1.3, 1.4, 1.6) , Chapter 2(2.2, 2.3, 2.4, 2.5)**	8
	PRACTICAL: 1) Design of ALU for a CPU 2) Design of a simple computer with basic inputs and outputs 3) Design of memory logic for computer interface	
	RESEARCH: 1) VLSI design of a CPU using CMOS logic 2) VLSI testing and Diagnosis 3) Design of simple memory using static CMOS circuits 4) Design of memory using Dynamic CMOS logic	
4	**Input/output Organization:** Accessing I/O Devices, Interrupts – Interrupt Hardware, Enabling and Disabling Interrupts, Handling Multiple Devices **Direct Memory Access**: Bus Arbitration, Speed, size and Cost of memory systems. Cache Memories – Mapping Functions. **Text book 2: Chapter 4 (4.1, 4.2.1, 4.2.2, 4.2.3, 4.4), Chapter 5 (5.4, 5.5.1)**	9
	PRACTICAL: 1) Device driver development 2) Programmed I/O vs Interrupt-Driven I/O Performance Comparison 3) DMA (Direct Memory Access) Implementation and Analysis	
	RESEARCH: 1) AI-Based Interrupt Load Balancing in Multicore Systems 2) Secure Interrupt Handling Against Hardware Attacks 3) GPU Interrupt Handling Optimization	

5	**Basic Processing Unit:** Some Fundamental Concepts: Register Transfers, Performing ALU operations, fetching a word from Memory, Storing a word in memory. Execution of a Complete Instruction. **Pipelining:** Basic concepts, Role of Cache memory, Pipeline Performance. **Text book 2: Chapter 7 (7.1, 7.2), Chapter 8(8.1)**	8
	PRACTICAL: 1) Simulation of Instruction Execution Cycle 2) Register Transfer Operations Implementation 3) Instruction Pipeline Implementation 4) Branch Instruction Execution in CPUs 5) Hardwired vs Microprogrammed Control Unit Comparison	
	RESEARCH: 1) AI-Assisted Processor Optimization 2) Low-Power Processor Design for Mobile Devices 3) Fault-Tolerant CPU Design	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Digital Design with an Introduction to Verilog Design	M. Morris Mano & Michael D. Ciletti	6th	Pearson Education	2024
2	Computer Organization	Carl Hamacher, Zvonko Vranesic, SafwatZaky	6th	Tata McGraw Hill.	2023

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Computer Organization & Architecture.	William Stallings	9th	Pearson Education	2014
2	Fundamentals of Digital Logic with Verilog design	Stephen Brown, Zvonko Vranesic	1st	McGraw- Hill	2009
3	Fundamentals of HDL.	Cyril Prasanna Raj	1st	Pearson Education	2009

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Web-Based Interactive Educational Resource**	https://cse11-iiith.vlabs.ac.in/
2	**NPTEL**	https://archive.nptel.ac.in/courses/106/105/106105165/
3	**NPTEL**	https://youtu.be/vBURTt97EkA
4	**YouTube**	https://www.youtube.com/playlist?list=PLxCzCOWd7aiGmXg4NoX6R31AsC5LeCPHe

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply the K–Map techniques to simplify various Boolean expressions.
CO2	Design different types of combinational and sequential circuits.
CO3	Describe the fundamentals of machine instructions, addressing modes and Processor performance.
CO4	Explain the approaches involved in achieving communication between processor and I/O devices.
CO5	Analyse internal Organization of Memory and Impact of cache/Pipelining on Processor Performance.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	-	-	-	-	-	-	-	1	2	-	-
CO2	3	2	2	-		-	-	-	-	-	-	2	-	-
CO3	2	2	2	-	-	-	-	-	-	-	-	1	-	-
CO4	2	2	2	-	-	-	-	-	-	-	2	1	-	-
CO5	2	2	2	-	-	-	-	-	-	-	2	1	-	-
TOTAL	12	10	10	-		-	-	-	-	-	5	7	-	-
AVG	2.4	2	2	-		-	-	-	-	-	1.6	1.4	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes:

- Chalk and Talk with Black Board.
- ICT based Teaching.
- Demonstration based Teaching.
- Adopt Problem Based Learning (PBL), which fosters students‟ Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
- Demonstrate HDL tools like VERILOG to implement Combinational and Sequential Circuits.

DATA STRUCTURES LABORATORY

Course Code	B25CSL36	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	PCCL	**Total SAAE Hours / semester**	02	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9, and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. Linear data structures and their applications such as stacks, queues and list 2. Non-Linear data structures and their applications such as trees and graphs

PREREQUISITES
1. Knowledge of relevant Data Structures

Sl. No.	LIST OF PROGRAMS
1	Develop a C program to sort the elements of an array using Radix sort technique.
2	Develop a C Program to implement Hashing using Linear and Quadratic Probing.
3	Develop a menu driven program in C to implement primitive operations on stack a) Push b) Pop c) Display. The program should print appropriate messages for stack overflow and stack underflow.
4	Develop a C program to convert INFIX notation to POSTFIX notation.
5	Develop a C program for evaluation of POSTFIX notation.
6	Develop a menu driven program in C language to implement QUEUE data structures that performs following primitive operations: a) insert b) delete and c) display.
7	Develop a menu driven program in C language to for CIRCULAR QUEUE that performs following primitive operations: a) insert b) delete and c) display.
8	Develop a menu driven program in C to perform primitive operations on single linked list.
9	Develop a C program to perform insertion, deletion, and traversal of a binary search tree.
10	Develop a C program to traverse a graph using Breadth First Search a C program to traverse a graph using Breadth First Search.

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Develop programs related to arrays and hashing.
CO2	Develop programs related to stack and its applications.
CO3	Develop programs related to queues.
CO4	Demonstrate use of dynamic memory allocation to implement single linked list.
CO5	Develop programs related to trees.
CO6	Apply concept of graphs to implement traversal algorithms.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	3	-	-	3	-	-	2	3	3	-
CO2	3	3	3	-	3	-	-	3	-	-	2	3	3	-
CO3	3	3	3	-	3	-	-	3	-	-	2	3	3	-
CO4	3	3	3	-	3	-	-	3	-	-	2	3	3	-
CO5	3	3	3	-	3	-	-	3	-	-	2	3	3	-
CO6	3	3	3	-	3	-	-	3	-	-	2	3	3	-
TOTAL	18	18	18	-	18	-	-	18	-	-	12	18	18	-
AVG	3	3	3	-	3	-	-	3	-	-	2	3	3	-

DATA ANALYTICS WITH EXCEL

Course Code	B25CSL371	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE Hours / semester**	02	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. Develop proficiency in using Excel for performing basic to advanced data operations, functions, and formulae. 2. Enable students to analyse and visualize data effectively using various charting tools and pivot tables. 3. Apply logical and conditional functions to solve real-life decision-making problems in data. 4. Explore advanced data handling and analysis features such as lookup functions, conditional formatting, scenario analysis, and data referencing.

PREREQUISITES
1. Analytical Thinking 2. Mathematical Reasoning

Sl. No.	LIST OF PROGRAMS
1	Perform the following basic EXCEL functions. a) Mathematical operations with functions (SUM, COUNT, MAXIMUM, ROUND) and without functions b) Logical functions like IF, AND, OR, NOT c) Mischalleneous functions like RAND, ISBLANK, COUNTIF) Additional Excel tricks (paste special -leaving blank fields, transpose rows & columns and paste)
2	Create cells with different Data Formats and apply all the conditional formatting options.
3	Consider Employee Data which consists of the fields- Employee id, Department Name, Employee Name, Date of Joining. Perform advanced data sorting including a) Filtering options b) Date filtering c) Filtering into different sheet d) Custom sorting
4	Consider Students data with the fields Name, Test1, Test2, Test3 and Total. a) Highlight the students who have scored more than 80 in Test2 b) Display Students who scored below average marks in Test1 c) Show Data Bars for Test2 d) Insert different icons for i)total >200 ii)total between 180 & 200 iii)total<180

	e) For students who missed 2 or more test than assign a D and show alert icon
5	Consider Personal Data with the following fields – First Name, Last Name, DOB, State Zip, Description, and Region. Apply the following text formulas: a) Merge First Name and Last Name into Full Name b) Use SEARCH and LEFT/MID to extract specific substrings c) Use RIGHT and LEN to extract Last Name from Full Name d) Format date using the DATE function e) Use MID and - to extract patterns (e.g., from 'California/west:276', extract 'west') f) Demonstrate Absolute, Relative, and Mixed Cell Referencing
6	Consider a Sales Data with the following fields Month, sales- Product1, sales-product2. a) Perform a VLOOKUP, HLOOKUP & XLOOKUP operations (E.g- lookup product1 sales and copy in another table)
7	Consider an employee sales data in several months. a) Apply INDEX and MATCH formulas to lookup sales values of employees in different months. b) Apply Text to column Field to delimit with special characters, custom attern. c) Perform nested MATCH function and other advance functions.
8	Consider a customer sales order (large data set). Use Pivot table to a) Check total sales by region b) Filter data to focus on specific products/order method c) Use various charts to visualize the data
9	Consider a Retail Sales dataset containing Order ID, Date, Region, Product Category, Sales Amount, Quantity, and Sales Representative. Create an interactive dashboard to provide managerial insights: a) Use Pivot Charts, Slicers, and Timeline Filters b) Show total sales by Region, Top Products, and monthly trends c) Arrange all components in a single dashboard view
10	Perform What-If Analysis using PMT formula and Scenario Manager on loan details: a) Analyse how varying Loan Amount, Interest Rate, and Time Period affect monthly EMI b) Create at least three scenarios and compare results visually
11	Consider a raw Excel file with quarterly sales data from different branches. The data includes merged cells, inconsistent date formats, and null values. Use Power Query to restructure and clean the data: a) Remove merged cells and fill down missing values b) Standardize date formats c) Unpivot quarterly columns for normalized analysis d) Merge/append tables if needed
12	Consider a monthly sales report containing sales data of different products. The file includes columns such as Product Name, Sales Amount, Region, and Month. Automate formatting using Macros. a) Highlight sales above ₹50,000 using conditional formatting b) Adjust column widths for all columns c) Insert a summary table to calculate Total and Average sales per report d) Record the macro and assign it to a command button for reuse. e) View and execute the generated VBA code. f) Create a simple message box macro (MsgBox "Report Generated")

PRESCRIBED MATERIAL

Sl. #	Type of E-Resource	**URL** TechA Microsoft Excel for Data Analysts Certification
1	Online course	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013837497517228032590_shared/overview

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply Excel formulas, functions, formatting, and logical operations.
CO2	Analyse and manipulate datasets using advanced Excel tools.
CO3	Create dashboards and perform decision-support analytics.
CO4	Automate workflows and transform data using Power Query and Macros.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	1	-	2	-	-	-	2	1	2	3	1	-
CO2	2	2	1	-	2	-	-	-	2	1	2	3	1	-
CO3	2	2	1	-	2	-	-	-	2	1	2	3	1	-
CO4	2	2	1	-	2	-	-	-	2	1	2	3	1	-
TOTAL	8	8	4	-	8	-	-	-	8	4	8	12	4	-
AVG	2	2	1	-	2	-	-	-	2	1	2	3	1	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. ICT based Teaching.
2. Demonstration based Teaching.

R PROGRAMMING

Course Code	B25ISL372	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE hours / semester**	02	**SEE MARKS**	50
L: T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4,9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable cities & Communities		

COURSE LEARNING OBJECTIVES
1. To explore and understand how R and R Studio interactive environment. 2. To understand the different data Structures, data types in R. 3. To learn and practice programming techniques using R programming. 4. To import data into R from various data sources and generate visualizations. 5. To draw insights from datasets using data analytics techniques

PREREQUISITES
1. Programming Concepts 2. Basic Mathematics & Statistics knowledge

Sl. No.	**LIST OF PROGRAMS**
1	Demonstrate the installation of the R-Programming environment and illustrate the use of install. Packages() command in R.
2	Write an R program. Perform the following: a) Assign different type of values to variables and display the type of variable. Assign different types such as Double, Integer, Logical, Complex and Character and understand the difference between each data type. b) Demonstrate Arithmetic and Logical Operations with simple examples. c) Demonstrate generation of sequences and creation of vectors. d) Demonstrate Creation of Matrices e) Demonstrate the Creation of Matrices from Vectors using Binding Function. f) Demonstrate element extraction from vectors, matrices and arrays
3	Assess the Financial Statement of an Organization being supplied with 2 vectors of data: Monthly Revenue and Monthly Expenses for the Financial Year. You can create your own sample data vector for this experiment) Calculate the following financial metrics: a) Profit for each month. b) Profit after tax for each month (Tax Rate is 30%). c) Profit margin for each month equals to profit after tax divided by revenue. d) Good Months – where the profit after tax was greater than the mean for the year. e) Bad Months – where the profit after tax was less than the mean for the year. Note:

<table>
<tr><td></td><td>a) All Results need to be presented as vectors
b) Results for Dollar values need to be calculated with $0.01 precision, but need to be presented in Units of $1000 (i.e. 1k) with no decimal points
c) Results for the profit margin ratio need to be presented in units of % with no decimal point.
d) It is okay for tax to be negative for any given month (deferred tax asset)
e) Generate CSV file for the data</td></tr>
<tr><td>4</td><td>Develop a program to create two 3 X 3 matrices A and B and perform the following operations a) Transpose of the matrix b) addition c) subtraction d) multiplication</td></tr>
<tr><td>5</td><td>Develop a program to find the factorial of given number using recursive function calls</td></tr>
<tr><td>6</td><td>Develop an R Program using functions to find all the prime numbers up to a specified number by the method of Sieve of Eratosthenes.</td></tr>
<tr><td>7</td><td>The built-in data set mammals contain data on body weight versus brain weight. Develop R commands to:
a) Find the Pearson and Spearman correlation coefficients. Are they similar?
b) Plot the data using the plot command.
c) Plot the logarithm (log) of each variable and see if that makes a difference.</td></tr>
<tr><td>8</td><td>Develop R program to create a Data Frame with following details and do the following operations.
<table>
<tr><th>itemCode</th><th>itemCategory</th><th>itemPrice</th></tr>
<tr><td>1001</td><td>Electronics</td><td>700</td></tr>
<tr><td>1002</td><td>Desktop Supplies</td><td>300</td></tr>
<tr><td>1003</td><td>Office Supplies</td><td>350</td></tr>
<tr><td>1004</td><td>USB</td><td>400</td></tr>
<tr><td>1005</td><td>CD Drive</td><td>800</td></tr>
</table>
a) Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.
b) Subset the Data frame and display only the items where the category is either “Office Supplies” or “Desktop Supplies”
c) Create another Data Frame called “item-details” with three different fields item Code, Item QtyonHand and Item ReorderLvl and merge the two frames</td></tr>
<tr><td>9</td><td>a) Let us use the built-in dataset air quality which has Daily air quality measurements in New York, May to September 1973. Develop R program to generate histogram by using appropriate arguments for the following statements.
b) Assigning names, using the air quality data set.
c) Change colors of the Histogram
d) Remove Axis and Add labels to Histogram
e) Change Axis limits of a Histogram
f) Add Density curve to the histogram</td></tr>
<tr><td>10</td><td>Create pie charts and bar charts in R for data visualization.</td></tr>
</table>

TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Scientific Programming and Simulation Using R	Jones, O., Maillardet. R. and Robinson, A.	2nd	Chapman & Hall/CRC, The R Series.	2014
2	Learning R: A Step by Step Function Guide to Data Analysis	Richard cotton	1st	O’Reilly Media Inc.	2013

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**NPTEL**	https://onlinecourses.nptel.ac.in/e-learning/preview/noc24_ma95
2	**Geeksfor geeks**	https://www.geeksforgeeks.org/r-language/r-tutorial/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Understand the fundamental syntax of R data types, expressions and the usage of the R-Studio IDE
CO2	Apply R Programming constructs: conditionals, looping, Arrays, lists and functions to solve problems
CO3	Analyse the data using visualization packages and file handlers.
CO4	Develop a solution to solve real time problems using R Programming language.
CO5	Evaluate a real time Data Set for a specific application using R Programming.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	-	-	3	-	-	-	-	1	-	2	2	-
CO2	2	2	2	-	3	-	-	-	-	1	-	3	2	-
CO3	2	3	2	2	3	-	-	-	-	1	-	3	3	-
CO4	2	3	3	2	3	-	-	2	2	1	-	3	3	-
CO5	2	3	3	3	3	-	-	2	2	2	-	3	3	-
TOTAL	10	12	10	7	15	-	-	4	4	6	-	14	13	-
AVG	2	2.4	2.5	2.3	3	-	-	2	2	1.2	-	2.8	2.6	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.
1) Chalk and Talk with Black Board.
2) ICT based Teaching.
3) Demonstration based Teaching.

PROJECT MANAGEMENT WITH GIT

Course Code	B25ADL373	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE Hours / semester**	2	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation and Infrastructure **SDG 11**: Sustainable Cities and Communities:		

COURSE LEARNING OBJECTIVES
1. To familiar with basic command of Git. 2. To create and manage branches. 3. To understand how to collaborate and work with Remote Repositories. 4. To familiar with virion controlling commands.

PREREQUISITES
1.Basic Terminal Commands 2.Operating System(Windows/Linux)

Sl. No.	LIST OF PROGRAMS
1	**Setting up and Basic Commands** Initialize a new Git repository in a directory. Create a new file and add it to the staging area and commit the changes with an appropriate commit message.
2	**Creating and Managing Branches** Create a new branch named "feature-branch." Switch to the "master" branch. Merge the "feature-branch" into "master."
3	**Creating and Managing Branches** Write the commands to stash your changes, switch branches, and then apply the stashed changes.
4	**Collaboration and Remote Repositories** Clone a remote Git repository to your local machine.
5	**Collaboration and Remote Repositories** Fetch the latest changes from a remote repository and rebase your local branch onto the updated remote branch.
6	**Collaboration and Remote Repositories** Write the command to merge "feature-branch" into "master" while providing a custom commit message for the merge.
7	**Git Tags and Releases** Write the command to create a lightweight Git tag named "v1.0" for a commit in your local repository.
8	**Advanced Git Operations** Write the command to cherry-pick a range of commits from "source-branch" to the current branch.

9	**Analysing and Changing Git History** Given a commit ID, how would you use Git to view the details of that specific commit, including the author, date, and commit message?
10	**Analysing and Changing Git History** Write the command to list all commits made by the author "JohnDoe" between "2023-01-01" and "2023-12-31."
11	**Analysing and Changing Git History** Write the command to display the last five commits in the repository's history.
12	**Analysing and Changing Git History** Write the command to undo the changes introduced by the commit with the ID "abc123".

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Version Control with Git: Powerful Tools and Techniques for Collaborative Software Development	Prem Kumar Ponuthorai, Jon Loeliger	3rd	O'Reilly Media, Inc.	2022

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	Infosys Springboard	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01309444 33473699842782_shared/overview
2	Reference	https://git-scm.com/book/en/v2

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Use the basics commands related to git repository.
CO2	Create and manage the branches.
CO3	Apply commands related to Collaboration and Remote Repositories.
CO4	Use the commands related to Git Tags, Releases and advanced git operations.
CO5	Analyse and change the git history.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	-	-	3	-	-	-	-	-	-	2	3	-
CO2	2	3	2	-	3	-	-	-	1	-	-	2	3	-
CO3	2	2	2	1	3	-	-	-	3	3	-	2	3	-
CO4	2	3	2	2	3	-	-	-	1	-	1	2	3	-
CO5	3	4	-	2	2	-	-	-	-	-	1	2	3	-
TOTAL	12	13	6	5	14	-	-	-	5	3	2	10	15	-
AVG	2.4	2.6	2	1.6	2.8	-	-	-	1.6	3	1	2	3	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. Hands-on learning on Git Operations such as repository creation, branching, merging.
2. Demonstration-based teaching of Git commands.
3. Collaborative learning using group-based using local and remote repositories to understand distributed version control.

UNIX SHELL SCRIPTING

Course Code	B25CSL374	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE Hours / semester**	28	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	1			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry Innovation & Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. Understand the features, architecture of UNIX and its commands.
2. Discuss different UNIX files, attributes and permissions
3. Discuss filter programs and regular expressions
4. Understand essential facets of shell programming in order solve the shell script problems

PREREQUISITES
1. Basics of programming

Sl. No.	**LIST OF PROGRAMS**
1	Working on general purpose commands and General-Purpose Utilities
2	Working on general purpose commands and General-Purpose Utilities
3	Working on Ordinary files and Attributes
4	Working on Ordinary files and Attributes
5	Working on Filter programs and filters using regular expression
6	Working on Filter programs and filters using regular expression
7	a) Write a shell script which displays a list of all the files in the current directory to which you have read, write and execute permissions. b) Write a shell script which will accept a filename and starting and ending line numbers and displays these lines from given file. c) Write a shell script which is expected to accept two filenames as its arguments. Check the number of arguments and display the contents of the argument files if the arguments is two, otherwise display an error message an exit.
8	a) Write a shell program that will do the following tasks in order: i) clear the screen ii) print the current directory iii) display current login users iv) list of users in the system v) list of processes vi) list of files vii) today's date viii) Quit to UNIX b) Write a shell script that reads data from a text file and appends the lines starting with an alphabet from 'a' through 'k' to a file named "ak" and the lines from 'l' through 'r' to a file named "lr" and

	the lines from 's' through 'z' to a file named "sz". Then display the number of lines in each of the files ak, lr and sz individually. c) Write a shell script which will receive login name during execution, obtain information about it from /etc/passwd and display this information on screen in easily understandable format.
9	a) A shell script receives even number of filenames as arguments. Suppose four files are supplied as arguments then the first file should get copied into second, third file into fourth and so on. If odd number of filenames is supplied then no copying should take place and an error message should be displayed. b) Write a shell script which will receive any number of filenames as arguments. The shell script should check whether every argument supplied is a file or a directory. If it a directory it should be appropriately reported. If it is a filename then name of the file as well as the number of lines present in it should be reported.
10	a) Write a shell script which should display the message, "Good Morning", "Good Afternoon", "Good Evening", depending upon the time. b) Write a shell script which accepts any number of arguments and prints them in reverse order. Ex: If file name is test then $ sh test A B C should produce C B A. c) Write script average which computes the average value (rounded to an integer as computed with expr) of the numeric values given in the parameters and writes the result to standard output. For example average 4 13 112 7 outputs 34.

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Explain the fundamental concepts of UNIX Operating system along with the working of various commands.
CO2	Illustrate various filters to solve variety of applications.
CO3	Write Regular expressions for pattern matching.
CO4	Write shell scripts.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	3	-	-	-	-	-	2	1	2	-
CO2	3	3	2	-	3	-	-	-	-	-	2	1	2	-
CO3	3	3	2	-	3	-	-	-	-	-	2	1	2	-
CO4	3	3	2	-	3	-	-	-	-	-	2	1	2	-
TOTAL	12	12	8	-	12	-	-	-	-	-	8	4	8	-
AVG	3	3	2	-	3	-	-	-	-	-	2	1	2	-

Teaching-Learning Process

Pedagogy (General Instructions):

A successful pedagogy for Unix shell scripting focuses on practical application, gradual complexity, and real-world scenarios. This approach involves teaching fundamental concepts, then moving to more advanced techniques and scripting best practices. It also emphasizes hands-on experience and real-world problem-solving to solidify learning. Have a look at how the course can be made more interesting:

Practical Learning

1. **Hands-on Exercises:** Encourage students to practice writing scripts to solve real-world problems.
2. **Project-Based Learning:** Assign projects that involve creating scripts for specific tasks, such as automating backups, managing users, or monitoring system resources.
3. **Debugging Techniques:** Teach students how to identify and fix errors in their scripts using tools like set -x for tracing execution and error messages.

Resources and Tools:

1) **Online Courses and Tutorials:** Few links are given for references in the E-Resources table above.
2) **Documentation:** Recommend using the official Unix documentation (e.g., man pages) and online tutorials for reference.

Community Resources: Encourage students to participate in online forums and communities where they can ask questions and share knowledge.

ADDITIONAL MATHEMATICS I
Common to all branches

Course Code	B25DIP310	**Total contact hours / week**	02	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	20	**SEE MARKS**	-
L:T:P:S	0:0:0:0	**Total Notional Learning Hours**	48	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	4, 8 and 9	**SDG Initiatives**	**SDG4**: Quality Education **SDG 8**: Decent work and Economic Growth **SDG 9**: Industry, Innovation, and Infrastructure		

COURSE LEARNING OBJECTIVES
1. Understand the concepts of complex numbers and differentiation. 2. Familiarize the importance of Integral calculus.

PREREQUISITES
1. Basics of differentiation and integration

MODULE	TOPICS	Hours
1	**COMPLEX NUMBERS** Definitions, complex numbers as an ordered pair, real and imaginary parts, modulus and amplitude of a complex number, equality of a complex number, addition, subtraction, multiplication and division of complex numbers, polar form, Argand diagram, exponential form, expressing in the form of a±ib, problems. Text book 1: **Chapter** 19 (19.1-19.8)	6
2	**DIFFERENTIAL CALCULUS** Polar curves, angle between radius vector and tangent, Taylor's series, Maclaurin's series of simple functions for single variable. Text book 1: **Chapter** 4 (4.4, 4.7)	5
3	**PARTIAL DIFFERENTIATION** Definitions, Euler's theorem, total differentiation, Differentiation of composite functions. Jacobians-Illustrative examples and problems. Text book 1: **Chapter** 5 (5.2, 5.5, 5.7)	5
4	**DIFFERENTIAL EQUATIONS** Solution of first order and first degree differential equations – Variable separable method, Linear differential equations, Bernoulli's differential equations, exact differential equations. Text book 1: **Chapter** 11 (11.1-11.6, 11.9-11.11)	6
5	**INTEGRAL CALCULUS** Reduction formulae for $sin^n x, cos^n x$(with proof) examples. Double and Triple integrals-simple problems with standard limits. Beta and Gamma functions, relationship between Beta and Gamma functions-simple problems. Text book 1: **Chapter** 6 (6.1-6.3) **Chapter** 7 (7.1,7.5,7.15,7.16)	6

PRESCRIBED TEXT BOOKS

Text Book	Book Title	Authors	Edition	Publisher	Year
1	Higher Engineering Mathematics	B.S Grewal	43rd	Khanna Publishers	2015
2	Higher Engineering Mathematics	H.K Dass	10th	John Wiley and sons	2016

REFERENCE BOOKS

Reference Book	Book Title	Authors	Edition	Publisher	Year
1	Higher Engineering Mathematics	B. V. Ramana	11th	McGraw-Hill Education	2011
2	A Text book of Engineering Mathematics	N.P.Bali and Manish Goyal	10th	Laxmi Publications	2022
3	Engineering Mathematics - II	Pandurangappa C	2nd	Sanguine Technical Publications	2022

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**NPTEL**	https://nptel.ac.in/courses/111105160
2	**NPTEL**	https://nptel.ac.in/courses/127106019
3	**MOOCs**	https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-spring-2019/
4	**MOOCs**	https://ocw.mit.edu/courses/18-330-introduction-to-numerical-analysis-spring- 2012/pages/syllabus/
5	**VTU e- learning**	http://elearning.vtu.ac.in

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply the concepts of complex numbers including algebraic operations, polar form and exponential representation to the problems arising in engineering.
CO2	Evaluate Taylor's series, Maclaurin's series, and angle between radius vector and tangent for the polar curves..
CO3	Apply partial derivatives, total differentiation, and compute Jacobians for multivariable functions in science and engineering
CO4	Apply first order differential equations to problems arising in engineering.
CO5	Evaluate double and triple integrals and problems on beta and gamma functions.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO2	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO3	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO4	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO5	3	3	-	2	-	-	-	2	2	-	2	2	-	-
TOTAL	15	15	-	10	-	-	-	10	10	-	10	10	-	-
AVG	3	3	-	2	-	-	-	2	2	-	2	2	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the delivered lessons shall develop students' theoretical and applied mathematical skills.
2. State the need for Mathematics with Engineering Studies and provide real-life examples.
3. Support and guide the students for self–study.
4. You will also be responsible for assigning homework, grading assignments and quizzes, and documenting students' progress.
5. Encourage the students to group learning to improve their creative and analytical skills.
6. Show short-related video lectures in the following ways:
    - As an introduction to new topics (pre-lecture activity).
    - As a revision of topics (post-lecture activity).
    - As additional examples (post-lecture activity).
    - As an additional material of challenging topics (pre-and post-lecture activity)
    - As a model solution of some exercises (post-lecture activity).

PERSONALITY DEVELOPMENT

Course Code	B25PDK311	**Total contact hours / week**	1	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	14	**SEE MARKS**	NA
L:T:P:S	1:0:0:1	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	NA
Mapping to SDG#	4	**SDG Initiatives**	**SDG 4**: Quality Education		

COURSE LEARNING OBJECTIVES
1. Develop self-awareness and understand their personality strengths. 2. Enhance communication skills required for employability. 3. Build emotional intelligence and interpersonal effectiveness. 4. Demonstrate teamwork, collaboration and leadership competencies. 5. Exhibit professional excellence and workplace readiness behaviours. 6. Prepare effectively for campus recruitment processes and placements.

PREREQUISITES
1. Enrolment in the 3rd Semester of a B.E./B.Tech. program. 2. Basic proficiency in English (reading, writing, listening, and speaking). 3. Willingness to participate in discussions, presentations, and group activities. 4. Openness to learning and self-improvement. 5. Positive attitude toward professional development. 6. Willingness to develop communication, leadership, and interpersonal skills. 7. Basic computer literacy (MS Office, email, online meeting platforms). 8. Interest in future internships, placements, and career planning.

Module	**Topic**	**Mode of Delivery**	**Hours**
1	Self-Awareness and Personality Assessment	Classroom & Activity	4
2	Communication Skills for Employability	Classroom & Activity	4
3	Emotional Intelligence and Interpersonal Skills	Classroom & Activity	4
4	Teamwork and Leadership Development	Classroom & Activity	4
5	Professional Excellence and Workplace Readiness	Classroom & Activity	4
6	Placement Readiness Foundation	Classroom & Activity	4
		Total	**24**

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Author(s)	Edition	Publisher	Year
1	Personality Development and Soft Skills	Barun K. Mitra	2nd	Oxford University Press India	2016
2	Soft Skills – The 7 Habits of Highly Effective People	Stephen R. Covey	30th	Simon & Schuster	2020
3	You Can Win	Shiv Khera	Revised Edition	Bloomsbury India	2018

REFERENCE BOOKS

Reference Book #	Book Title	Author(s)	Edition	Publisher	Year
1	Business Communication Today	Courtland L. Bovee & John V. Thill	14th	Pearson Education	2019
2	Effective Technical Communication	M. Ashraf Rizvi	2nd	McGraw Hill Education (India)	2017

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	NPTEL Soft Skills Courses	nptel.ac.in/courses?utm_source=chatgpt.com
2	SWAYAM Learning Platform	Swayam Central
3	Mind Tools Personal Development Resources	Leadership & Management Learning Solutions For Leaders & Managers - Mindtools Membership

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Analyse personal strengths, weaknesses, values and career aspirations through self-awareness tools.
CO2	Demonstrate effective verbal, non-verbal and written communication skills for professional environments.
CO3	Apply emotional intelligence principles to manage self and build positive workplace relationships.
CO4	Work effectively in teams and demonstrate leadership competencies in organizational settings.
CO5	Exhibit professional behaviour, workplace etiquette and corporate readiness skills.
CO6	Successfully prepare for aptitude tests, interviews, group discussions and placement processes.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	-	-	-	-	-	3	-	2	-	3	-	-	3
CO2	-	-	-	-	-	-	2	2	3	-	2	-	-	3
CO3	-	-	-	-	-	-	3	3	2	-	2	-	-	3
CO4	-	-	-	-	-	-	2	3	2	2	2	-	-	3
CO5	-	-	-	-	-	-	3	2	3	2	2	-	-	3
CO6	2	2	-	-	-	-	2	2	3	-	2	-	-	3
TOTAL	4	2	-	-	-	-	12	12	15	4	13	-	-	18
AVG	2	2	-	-	-	-	2	2.4	2.5	2	2.16	-	-	3

Teaching-Learning Process

Pedagogy (General Instructions):

1. Interactive Lectures
- Concept introduction through short lectures and multimedia presentations.
- Facilitated discussions on personality, emotional intelligence, leadership, and professional behaviour.
- Real-world examples and workplace scenarios.

2. Self-Assessment and Reflection Activities
- Personality profiling and self-awareness assessments.
- SWOT analysis (Strengths, Weaknesses, Opportunities, Threats).
- Reflective journals and personal development plans.
- Goal-setting exercises.

3. Technology-Enabled Learning
- Learning through MOOCs (NPTEL, SWAYAM, Coursera, etc.).
- Online quizzes, assessments, and discussion forums.
- Video-based learning and self-paced modules.

4. Industry Interaction
- Expert talks by industry professionals, alumni, and HR managers.
- Panel discussions on employability expectations.
- Career guidance and mentoring sessions.

Non Credit Mandatory Course (NCMC)
Assessment Details (both CIE and SEE):
CIE: Each question carrying 1 Mark (No Negative Marking for wrong answers)
CIE-1: 50 Multiple Choice Questions – 60 Minutes - 50X1=50 Marks
CIE-2: 50 Multiple Choice Questions – 60 Minutes - 50X1=50 Marks
TOTAL= 100 Marks
SEE: Not Applicable

DISCRETE MATHEMATICAL STRUCTURES

Course Code	B25IS41	**Total contact hours / week**	04	**CIE MARKS**	50
Course Category	ASC	**Total SAAE Hours / semester**	48	**SEE MARKS**	50
L:T:P:S	3:0:0:0	**Total Notional Learning Hours**	90	**TOTAL MARKS**	100
Total credits	03			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. To understand discrete and continuous mathematical structures. 2. To impart basics of relations and functions. 3. To apply principles of Recurrence Relations to find the generating functions and solve the Recurrence relations. 4. To gain knowledge of groups and their properties to understand the importance of algebraic properties relative to various number systems.

PREREQUISITES
1. Basic set theory concepts. 2. Knowledge of propositional logic. 3. Concepts of permutations, and combinations. 4. Basic Mathematical foundation.

MODULE #	TOPICS	Hours
1	**Mathematical Logic-I:** Logical Connectives and Truth Tables, Tautology, Contradiction, Logical Equivalence, Rules of Inference. **Mathematical Logic -II:** Open statements, Logical Implications involving quantifiers, Methods of Proof and Methods of Disproof. **Textbook 1: Chapter 1(1.1, 1.2.1, 1.2.2, 1.2.4, 1.3), Chapter 2(2.1, 2.2, 2.4)**	9
	PRACTICAL: 1) Identify simple and compound propositions. 2) Determine whether a statement is tautology, Contradiction, Contingency, verify logical equivalence using truth tables	
	RESEARCH: 1) Research automated truth table generation methods. 2) Investigate applications of truth tables in artificial intelligence and expert systems.	
2	**Mathematical Induction and Recursive Definitions:** Mathematical Induction, Recursive Definitions. **Principles of Counting-I:** The Rules of Sum and Product, Permutations, Combinations, Binomial and Multinomial Theorems, Combinations with Repetition.	9

	Textbook 1: **Chapter 3(3.1, 3.2), Chapter 4 (4.1, 4.2, 4.3, 4.4, 4.5)**	
	PRACTICAL: 1) Verify mathematical identities using the principle of mathematical induction. 2) Calculate permutations with and without repetition. 3) Compute multinomial coefficients.	
	RESEARCH: 1) Research applications of induction in computer science algorithms. 2) Investigate induction in recursive data structures.	
3	**Relation and Functions:** Cartesian Products of Sets, Relations, Functions, The Pigeon- hole Principle, Composition of Functions, Invertible Functions. **Relations-II:** Zero-One Matrices and Directed Graphs, Properties of Relations, Equivalence Relations, Partial Orders **Textbook 1: Chapter 5(5.1, 5.2, 5.3, 5.4, 5.5, 5.6), Chapter 6(6.1, 6.2, 6.3, 6.4)**	8
	PRACTICAL: 1) Construct Cartesian products of finite sets. 2) Represent relations using Ordered pairs, tables, matrices, directed graphs	
	RESEARCH: 1) Investigate binary relations in artificial intelligence. 2) Research relational structures in graph theory.	
4	**Principles of Counting -II:** The Principle of Inclusion-Exclusion, Derangements, Rook Polynomials. **Recurrence Relations:** First-order Recurrence Relations, Second-order Homogeneous Recurrence Relations **Textbook 1: Chapter 7(7.1, 7.2, 7.3), Chapter 8(8.1, 8.2)**	8
	PRACTICAL: 1) Calculate the number of derangements for a given value of n and verify using the recursive formula 2) Find the number of elements in the union of multiple sets using the Inclusion–Exclusion Principle	
	RESEARCH: 1) Investigate overlapping dataset analysis in data science. 2) Research applications in database query optimization. 3) Explore counting methods in network systems	
5	**Graphs:** Graphs, Isomorphism, Subgroups, Operations on Graphs, Walks and their classification, Euler Circuits and Euler Trails. **Trees:** Trees and their Basic properties, Rooted Trees, Prefix codes and Weighted trees **Textbook 1: Chapter 9(9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.8), Chapter 10 (10.1, 10.2, 10.3)**	8
	PRACTICAL: 1) Implement graph representation using adjacency matrix and adjacency list. 2) Implement Huffman Coding using prefix codes.	
	RESEARCH: 1) Explore graph algorithms used in transportation and navigation systems. 2) Study Huffman coding and other prefix coding techniques for data compression.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Discrete Mathematical Structures	Dr. D.S.C.	5th	PRISM	2018
2	Discrete and Combinatorial Mathematics	Ralph P. Grimaldi	5th	Pearson Education	2004

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Discrete Mathematics – A Concept-based Approach	Basavaraj S Anami & Venakanna S Madalli	-	Universities Press	2016
2	Discrete Mathematics and its Applications	Kenneth H. Rosen,	6th	McGraw Hill	2007
3	A Treatise on Discrete Mathematical Structures	Jayant Ganguly	-	Sanguine-Pearson	2010
4	Discrete Mathematical Structures Theory and Applications	D.S. Malik and M.K. Sen	Latest Edition	Thomson	2004
5	Discrete Mathematics with Applications	Thomas Koshy	-	Elsevier Reprint	2008

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**VTU E-learning**	VTU e-Shikshana Program
2	**NPTEL Course**	https://nptel.ac.in/courses/106106094
3	**NPTEL Course**	https://onlinecourses.nptel.ac.in/e-learning/preview/noc22_cs49
4	**Tutorials Point**	https://www.tutorialspoint.com/discrete_mathematics/index.htm
5	**Geeksforgeeks**	https://www.geeksforgeeks.org/engineering-mathematics/discrete-mathematics-tutorial/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply concepts of logical reasoning and mathematical proof techniques in proving theorems and statements.
CO2	Apply principles of mathematical induction, recursive definitions, and counting techniques to solve discrete mathematical problems.
CO3	Apply the basic concepts of relations, functions and partially ordered sets for computer representations.
CO4	Apply the Principle of Inclusion–Exclusion, derangements, rook polynomials, and recurrence relations to solve combinatorial counting and mathematical problems.
CO5	Apply tree-based structures and graph concepts to model and solve real-world computational problems.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	1	2	-	-
CO2	3	3	3	-	-	-	-	-	-	-	1	2	-	-
CO3	3	3	3	-	-	-	-	-	-	-	1	2	-	-
CO4	3	3	3	-	-	-	-	-	-	-	1	2	-	-
CO5	3	3	3	-	-	-	-	-	-	-	1	2	-	-
TOTAL	15	15	15	-	-	-	-	-	-	-	5	10	-	-
AVG	3	3	3	-	-	-	-	-	-	-	1	2	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Chalk and Talk with Black Board
2) ICT based Teaching
3) Demonstration based Teaching

DATABASE MANAGEMENT SYSTEMS

Course Code	B25IS42	**Total contact hours / week**	05	**CIE MARKS**	50
Course Category	IPCC	**Total SAAE Hours / semester**	50	**SEE MARKS**	50
L:T:P:S	3:0:2:0	**Total Notional Learning Hours**	120	**TOTAL MARKS**	100
Total credits	04			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. To Understand the fundamental architecture and components of a DBMS. 2. To Design the Model data using Entity–Relationship (ER) diagrams. 3. To Illustrate the SQL fundamentals for database creation and querying. 4. To Apply relational algebra and normalization techniques. 5. To Understand the basic transaction and concurrency control concepts.

PREREQUISITES
1. Basics Knowledge on Database.

MODULE #	**TOPICS**	**Hours**
1	**Introduction to Databases**: Introduction, Characteristics of database approach, Advantages of using the DBMS approach, History of database applications. **Overview of Database Languages and Architectures**: Data Models, Schemas, and Instances. Three schema architecture and data independence, database languages, and interfaces, The Database System environment. **Conceptual Data Modelling using Entities and Relationships**: Entity types, Entity sets and structural constraints, Weak entity types, ER diagrams, Specialization and Generalization. **Textbook 1**: **Chapter 1(** 1.1-1.8,) **Chapter 2(**2.1-2.4), **Chapter 3(** 3.3-3.10)	9
	PRACTICAL: 1) Identification of real-world data requirements and modeling using ER diagrams. 2) Design of ER diagrams for sample applications such as library management, student information systems, and inventory management. 3) Conversion of ER diagrams into relational schemas.	
	RESEARCH: 1) Systematic analysis of file-based systems and DBMS-based systems in real-world applications. 2) Exploration of emerging database models.	

2	**Relational Model**: Relational Model Concepts, Relational Model Constraints and relational database schemas, Update operations, transactions, and dealing with constraint violations. **Relational Algebra:** Unary and Binary relational operations, additional relational operations (aggregate, grouping, etc.) Examples of Queries in relational algebra. **Mapping Conceptual Design into a Logical Design:** Relational Database Design using ER-to-Relational mapping. **Textbook 1**: **Chapter 5 (**5.1-5.3), **Chapter 8(**8.1-8.5), **Chapter 9(**9.1) **Textbook 2**: **Chapter 3(** 3.2,3.3)	9
	PRACTICAL: 1) Design relational schemas from given ER diagrams. 2) Apply relational algebra operations (selection, projection, join, division) on sample relations. 3) Solve query problems using relational algebra expressions.	
	RESEARCH: 1) Comparative study of relational algebra versus SQL query formulation. 2) Analysis of integrity constraints and their role in maintaining database consistency. 3) Study of limitations of the relational model and motivation for alternative data models.	
3	**SQL Fundamentals** SQL Data Definition Language (DDL), Creating tables, altering schema, Data Manipulation (INSERT, DELETE, UPDATE), SELECT queries, conditions, Aggregate functions, GROUP BY, HAVING, Joins: INNER, OUTER, Subqueries and nested queries and Views. **Textbook 1**: **Chapter 6(** 6.1-6.5), **Chapter 7** (7.1-7.4)	8
	PRACTICAL: 1) Create and manage database tables using SQL DDL commands. 2) Perform data insertion, deletion, and updates using DML commands. 3) Write SQL SELECT queries with conditions, joins, and aggregate functions. 4) Implement subqueries, views, and indexes on sample databases. 5) Execute SQL queries using tools such as SQL, MySQL or SQLite.	
	RESEARCH: 1) Comparative study of SQL versus NoSQL query languages. 2) Study of performance impact of joins and nested queries. 3) Exploration of real-world SQL usage in enterprise applications.	
4	**Database Design & Normalization** Functional dependencies, Normal forms: 1NF, 2NF, 3NF, BCNF, Anomalies in database design, Normalization algorithms, Design guidelines and examples. **Textbook 1**: **Chapter 14(** 14.1-14.7), **Chapter 15(** 15.1-15.6)	8
	PRACTICAL: 1) Identification of functional dependencies from given relational schemas. 2) Detection of insertion, deletion, and update anomalies. 3) Normalization of relations up to 3NF and BCNF.	
	RESEARCH: 1) Comparative study of normalization versus denormalization strategies. 2) Analysis of the impact of normalization on database performance.	

	3) Study of normalization challenges in large-scale and distributed databases.	
5	**Transactions and Concurrency** Transaction concepts and properties, ACID properties, Concurrency control basics, Locking mechanisms, Granularity of Data Items and Multiple Granularity Locking, Other Concurrency Control Issues. **Textbook 1**: **Chapter 20(** 20.1- 20.3), **Chapter 21(** 21.1, 21.2, 21.5, 21.7)	8
	PRACTICAL: 1) Demonstration of transaction execution using COMMIT and ROLLBACK. 2) Simulation of concurrent transactions and identification of anomalies (lost update, dirty read). 3) Observation of transaction isolation levels in relational databases.	
	RESEARCH: 1) Study of concurrency control techniques in modern DBMS. 2) Exploration of transaction management in distributed and cloud databases.	

PRACTICAL COMPONENT OF IPCC

Sl. No.	Programs
1	Consider the schema for Company Database: **EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo)** **DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)** **DLOCATION (DNo, DLoc)** **PROJECT (PNo, PName, PLocation, DNo)** **WORKS_ON (SSN, PNo, Hours)** **Basic Queries** Write a query to 1. List all employees in the company. 2. Display names and salaries of employees earning more than 50,000 and less than 75,000 3. Retrieve details of employees working in department number 5 and 8 also 4. List all female employees. 5. Find all projects located in a specific city. ◆ **Intermediate Queries** Write a query to 6. Display names of employees along with their department names. 7. List all employees of the project in an ascending order 8. Find the manager of each department 9. Find employees working on more than one project 10. Display the total number of employees in each department 11. Retrieve the maximum salary and average salary the among the employees. ◆ **Join-Based Queries** Write a query to 12. List employee names and project names they are working on. 13. Display employee names with the hours worked on each project. 14. Show department names along with their locations. 15. List employees supervised by a specific manager. 16. Display all departments and their managers' names. ◆ **Aggregation Queries** Write a query to 17. Count number of employees in each department. 18. Find the average salary in each department.

19. Calculate total salary paid in each department.
20. Find the number of projects in each department.
21. Display average working hours per project.

◆ **Insert Queries**

Write a query to

22. Insert a new employee record.
23. Add a new department.
24. Insert a new project.
25. Add a new department location.
26. Insert data into WORKS_ON table.

◆ **Update Queries**

Write a query to

27. Increase salary of all employees by 10%.
28. Update department number of a specific employee
29. Change location of a project.
30. Update working hours of an employee on a project.
31. Update manager of a department.

◆ **Delete Queries**

Write a query to

31. Delete an employee record.
32. Delete a project entry.
33. Remove a department.
34. Delete records where employees work fewer than defined hours.
35. Delete a department location.

◆ **View Queries**

Write a query to

36. Create a view showing employee names and salaries.
37. Create a view for employees earning above a certain salary.
38. Create a view showing project details.

◆ **Transaction Queries**

Write a query to

39. Demonstrate a transaction using COMMIT.
40. Demonstrate a transaction using ROLLBACK.

◆ **Trigger Queries**

Write a query to

41. Create a trigger to prevent insertion of negative salary.
42. Create a trigger to log employee insert operations.
43. Create a trigger to log employee deletions.

◆ **Assertion Queries**

44. Create an assertion to ensure salary is not negative.
45. Create an assertion ensuring every department has a manager.

◆ **Cursor Queries**

46. Write a cursor to display all employee names.
47. Write a cursor to update salaries of employees.
48. Write a cursor to count total employees.

◆ **Report Generation**

49. Generate a report showing employees, departments, and projects.
50. Generate a report summarizing department-wise employee count and salary details.

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Database System Concepts	R. Elmasri, S.B. Navathe	7th	Pearson	2016
2	Database Management Systems	R. Ramakrishnan, J. Gehrke	3rd	McGraw-Hill.	2003

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Database System Concepts	Abraham Silberschatz Henry F. Korth S. Sudarshan	7th	Mc GrawHill	2021
2	Database Systems: Design, Implementation, Management	Carlos Coronel, Steven Morris	14th	Course Technology Inc.	2022

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Swayam NPTEL**	https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs39
		https://nptel.ac.in/courses/106106220
		https://nptel.ac.in/courses/106104135

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Describe DBMS architecture and distinguish it from file systems, Construct ER diagrams for real-world scenarios.
CO2	Apply relational algebra operations in database querying.
CO3	Formulate SQL queries to perform CRUD operations.
CO4	Normalize a database up to 3NF to avoid redundancy and anomalies.
CO5	Understand basics of transactions, locking, and recovery concepts.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1	2	1	-	-	-	1	1	1	1	-	1	1
CO2	3	1	2	1	-	-	-	1	1	1	1	-	1	1
CO3	3	1	2	1	-	-	-	1	1	1	1	-	1	1
CO4	3	1	2	1	-	-	-	1	1	1	1	-	1	1
CO5	3	1	2	1	-	-	-	1	1	1	1	-	1	1
TOTAL	15	5	10	5	-	-	-	5	5	5	5	-	5	5
AVG	3	1	2	1	-	-	-	1	1	1	1	-	1	1

Teaching-Learning Process

Pedagogy (General Instructions):

1) Lectures with board / slides / online demonstrations.
2) Hands-on SQL examples using MySQL / SQLite.
3) Class assignments and group learning.

COMPUTER NETWORKS

Course Code	B25CS43	**Total contact hours / week**	05	**CIE Marks**	50
Course Category	IPCC	**Total SAAE Hours / semester**	50	**SEE Marks**	50
L:T:P:S	3:0:2:0	**Total Notional Learning Hours**	120	**Total Marks**	100
Total credits	04			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation and Infrastructure **SDG 11:** Sustainable Cities and Communities.		

COURSE LEARNING OBJECTIVES
1. Describe communication protocols and layered network architectures and introduce the fundamental various types of computer networks. 2. To demonstrate the TCP/IP and OSI models with merits and demerits and Explain convention computer system interfacing standards and peer to peer data link protocols. 3. To explore the various layers of OSI Model and to introduce UDP and TCP Models. 4. To obtain a thorough knowledge of Information and Network Security.

PREREQUISITES
1. An understanding of the basics of programming

MODULE #	TOPICS	Hours
1	Introduction: Data Communications, Networks, Network Types, Protocol Layering, TCP/IP Protocol suite, The OSI model, Physical Layer: Transmission media. **Textbook 1: Chapter** 1: (1.1 - 1.6), **Chapter** 2(2.6).	8
	PRACTICAL: 1) Analyze protocol encapsulation and decapsulation using Wireshark packet captures. 2) Observe different networking devices (Hub, Switch, Router, Repeater). Measure cable types and connectors.	
	RESEARCH: 1) Research the evolution of data communication from telegraph to 6G technology. 2) Research current Physical Layer technologies such as fiber optics, 5G, and high-speed Ethernet.	
2	Data Link Layer: Data link control: Framing, Error Control, Media Access Control: Random Access, Controlled Access. Network Layer: Services, Packet Switching, performance, Internet Protocol Version 4, Next Generation IP (IPV6). **Textbook1: Chapter** 3: (3.2,3.2.1, 3.2.2,3.3, 3.3.1, 3.3.2), **Chapter** 7: (7.1–7.3,7 .4-7.4.1,7.4.4,7.1, 7.5,7.5.1 ,7.5.2, 7.5.3).	9
	PRACTICAL: 1) Analyze Ethernet frames using Wireshark. Identify frame fields such as preamble, source MAC, destination MAC, type, data, and FCS. 2) Measure latency, throughput, and packet loss using Ping and Traceroute. Analyze results under different network conditions.	
	RESEARCH: 1) Research IEEE 802 standards and the role of the MAC sublayer in Ethernet and Wi-Fi.	

	2) Study Quality of Service (QoS) techniques and factors affecting network performance.	
3	Network Layer: Routing of Packets Introduction, Routing Algorithms-DVR, LSR, PVR, Unicast Routing Protocols: Unicast Routing protocols: RIP, OSPF, BGP4, Multicasting Routing-MOSPF. **Textbook 1: Chapter** 8: (8.1,8.2,8.3, 8.4.4).	8
	PRACTICAL: 1) Configure a simple routed network using Cisco Packet Tracer. Observe packet forwarding between multiple networks using ping and traceroute. 2) Study BGP configuration concepts using network simulators (Packet Tracer/GNS3) or analyze sample BGP routing tables.	
	RESEARCH: 1) Research OSPF architecture, areas, LSAs, and compare its performance with RIP. 2) Analyze RIP characteristics, limitations, hop-count metric, and practical applications.	
4	Transport Layer: Services, Transport-Layer Protocols: Introduction, User Datagram Protocol (UDP), Transmission Control Protocol: services, features, segments, TCP connections, state transition diagram, flow control, Error control, TCP Congestion control. **Textbook 1: Chapter** 9: (9.1, 9.2, 9.3,9.4, 9.4.1-9.4.5, 9.4.7-9.4.9).	9
	PRACTICAL: 1) Analyze end-to-end communication using Cisco Packet Tracer or Wireshark. Identify transport layer functions such as segmentation, multiplexing, and port addressing. 2) Observe the TCP three-way handshake and four-way connection termination using Wireshark.	
	RESEARCH: 1) Research TCP services, reliability mechanisms, and applications in web, email, and file transfer. 2) Research the role of the Transport Layer in reliable communication and compare it with the Network Layer.	
5	Application Layer: Introduction, Client-Server Programming, Standard Client Server Protocols: World Wide Web and HTTP, FTP, Electronic Mail, Domain Name System (DNS), TELNET, Secure Shell (SSH). **Textbook 1: Chapter** 10: (10.1-10.3).	8
	PRACTICAL: 1) Study the architecture of the Application Layer and identify commonly used application protocols using Wireshark. 2) Perform DNS lookups using tools such as nslookup, dig, or host. Analyze DNS query and response packets.	
	RESEARCH: 1) Study the evolution of HTTP (HTTP/1.1, HTTP/2, HTTP/3) and their performance improvements. 2) Compare Telnet and SSH with emphasis on authentication, encryption, and security.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Data Communication and Networking with TCP/IP Protocol Suite	Behrouz A. Forouzan	6th	McGraw-Hill Higher Education	2022

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Computer Networking	Kurose	8th	PEARSON	2022

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	Geeks for Geeks	https://www.geeksforgeeks.org/data-communication-definitioncomponents-types-channels
2	Tutorials point	https://www.tutorialspoint.com/data_communication_computer_network/index.html
3	NPTEL	https://nptel.ac.in/courses/106105081 Computer Networks

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Understand the basics of data communication, network models and physical layer.
CO2	Describe data link protocols, media access control methods, IPv4/IPv6 network addressing and packet-switching services to optimize the network performance.
CO3	Analyze the routing algorithms in network layer.
CO4	Explain transport layer services, features, flow control, and error control mechanisms of UDP and TCP using state transition diagrams.
CO5	Apply client-server programming principles to develop network applications and implement standard protocols to ensure secure and efficient web communication.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	-	-	-	-	-	-	1	1	3	3	-
CO2	3	3	2	2	1	-	-	-	-	1	1	3	3	-
CO3	3	3	2	2	2	-	-	-	-	1	1	3	3	-
CO4	2	2	2	-	-	-	-	-	-	1	1	3	3	-
CO5	3	2	2	1	-	-	-	-	-	1	1	3	3	-
TOTAL	14	13	8	5	3	-	-	-	-	5	5	15	15	-
AVG	2.8	2.6	2	1.6	1.5	-	-	-	-	1	1	3	3	-

Teaching-Learning Process Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Chalk and Talk with Black Board.
2) ICT based Teaching.
3) Demonstration based Teaching.

ANALYSIS AND DESIGN OF ALGORITHMS

Course Code	B25CS44	**Total contact hours / week**	03	**CIE MARKS**	50
Course Category	PCC	**Total SAAE Hours / semester**	48	**SEE MARKS**	50
L:T:P:S	3:0:0:0	**Total Notional Learning Hours**	90	**TOTAL MARKS**	100
Total credits	03			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. To learn the methods for analyzing algorithms and evaluating their performance. 2. To demonstrate the efficiency of algorithms using asymptotic notations. 3. To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound. 4. To learn the concepts of P and NP complexity classes.

PREREQUISITES
1. Programming in C. 2. Data Structures.

MODULE #	TOPICS	Hours
1	**INTRODUCTION:** What is an Algorithm? Fundamentals of Algorithmic Problem Solving. Important Problem types **FUNDAMENTALS OF THE ANALYSIS OF ALGORITHM EFFICIENCY:** Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical Analysis of Non recursive Algorithms, Mathematical Analysis of Recursive Algorithms. **Textbook 1; Chapter** 1(1.1-1.3), **Chapter** 2(2.1-2.4)	8
	PRACTICAL: 1) Implement recursive and iterative factorial computation. 2) Write a program to count basic operations in nested loops. 3) Write a program to calculate GCD using different methods.	
	RESEARCH: 1) Investigate algorithm efficiency challenges in large-scale data processing. 2) Comparative study of asymptotic analysis techniques used in real-world systems. 3) Research the role of computational complexity in cloud and distributed computing.	
2	**THE GREEDY METHOD:** Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Trees and codes **DECREASE-AND-CONQUER:** Algorithms for generating combinatorial objects-generating permutations, subsets. **Textbook 1: Chapter** 9 (9.1-9.4), **Chapter** 4 (4.3)	9
	PRACTICAL:	

	1) Write a program for activity selection using greedy method. 2) Generate permutations of a given string using recursion. 3) Implement basic Huffman encoding for a given set of characters and frequencies.	
	RESEARCH: Comparative study of greedy algorithms in networking and routing applications. Research applications of Huffman coding in data compression and multimedia transmission. Investigate combinatorial object generation techniques used in cryptography and AI. Explore recent graph optimization algorithms used in transportation and logistics. Research applications of Huffman coding in data compression and multimedia transmission.	
3	**DIVIDE AND CONQUER:** Multiplication of Large Integers and Strassen's Matrix Multiplication. **TRANSFORM-AND-CONQUER:** Balanced Search Trees – AVL Tree, 2-3 Tree, Heaps and Heapsort. **Textbook 1: Chapter** 5 (5.4) **Chapter** 6 (6.3, 6.4)	8
	PRACTICAL: 1) Implement merge sort for integer arrays. 2) Write a program for quick sort partition operation. 3) Construct a min heap using array representation.	
	RESEARCH: 1) Study advanced matrix multiplication algorithms beyond Strassen's method. 2) Research applications of balanced trees in database indexing and file systems. 3) Investigate heap structures in operating systems and priority scheduling. 4) Explore transform-and-conquer strategies in computational geometry. 5) Comparative analysis of self-balancing trees used in modern libraries and databases.	
4	**DYNAMIC PROGRAMMING:** Three basic examples (Coin Row Problem, Change Making Problem, Coin Collecting Problem). The Knapsack Problem and Memory Functions, Warshall's and Floyd's Algorithms. **Textbook 1; Chapter** 8 (8.1,8.2,8.4)	9
	PRACTICAL: 1) Write a program for Coin Row problem. 2) Write a program for Coin Making problem. 3) Write a program for Coin Collecting problem. 4) Implement Longest Common Subsequence (LCS).	
	RESEARCH: 1) Study dynamic programming applications in bioinformatics and genome sequencing. 2) Investigate optimization problems solved using memorization techniques. 3) Research shortest path algorithms in navigation and mapping systems. 4) Explore AI-based enhancements for dynamic programming optimization. 5) Comparative study of recursive, memorization, and tabulation techniques.	
5	**LIMITATIONS OF ALGORITHMIC POWER:** Decision Trees, P, NP, and NP-Complete Problems. **COPING WITH LIMITATIONS OF ALGORITHMIC POWER:** Backtracking (n-Queens problem, Subset-sum problem), Branch-and-Bound (Knapsack problem) **Textbook 1; Chapter** 11 (11.2, 11.3), **Chapter** 12 (12.1,12.2)	8
	PRACTICAL: 1) Write a program for graph coloring using backtracking. 2) Implement simple branch-and-bound for job assignment problem.	

	RESEARCH: 1) Study the significance of P, NP, and NP-Complete problems in computational theory. 2) Research practical applications of backtracking in puzzle solving and constraint satisfaction. 3) Investigate branch-and-bound techniques used in optimization software. 4) Survey heuristic and approximation methods for NP-hard problems. 5) Explore emerging quantum computing approaches for solving complex algorithmic problems.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Introduction to the Design and analysis of Algorithms	Anany Levitin	3rd	Pearson	2017

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein	4th	MIT	2022
2	Design and Analysis of Algorithms	S. Sridhar	2nd	Oxford	2023

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	E-book	https://homel.vsb.cz/~fai0013/Kniha_Algoritmy.pdf
2	Online course	Design and Analysis of Algorithms: https://nptel.ac.in/courses/106/101/106101060/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Explain the fundamental concepts of algorithms, problem-solving strategies, and evaluate algorithm efficiency using asymptotic notations and mathematical analysis.
CO2	Apply greedy technique and decrease and conquer method to solve graph based & combinatorial computational problems.
CO3	Demonstrate divide & conquer approaches and transform & conquer approaches to solve computational problems.
CO4	Make use of dynamic programming strategies to solve classical problems such as the coin row problem, change making, coin collecting, knapsack, and shortest path algorithms.
CO5	Discuss the limitations of algorithmic approaches through NP-completeness, and design solutions using backtracking and branch-and-bound methods.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO2	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO3	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO4	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO5	3	3	3	2	2	-	-	-	1	1	2	3	3	-
TOTAL	15	15	15	10	10	-	-	-	5	5	10	15	15	-
AVG	3	3	3	2	2	-	-	-	1	1	2	3	3	-

Teaching-Learning Process Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Chalk and Talk with Black Board.
2) ICT based Teaching.
3) Demonstration based Teaching.

FILE STRUCTURES

Course Code	B25IS45	**Total contact hours / week**	03	**CIE MARKS**	50
Course Category	PCC	**Total SAAE Hours / semester**	48	**SEE MARKS**	50
L:T:P:S	3:0:0:0	**Total Notional Learning Hours**	90	**TOTAL MARKS**	100
Total credits	03			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities:		

COURSE LEARNING OBJECTIVES
1. To provide an introduction to the fundamental file operations and storage systems. 2. To introduce fundamental concepts of file structure. 3. To familiarize the most important high-level file structures tools which include indexing, co-sequential processing. 4. To infer fast record access methods using B-Trees, B+ Trees. 5. To understand the Hashing techniques to file handling.

PREREQUISITES
1. Computer Organization / Operating Systems Basics. 2. Knowledge of RAM, ROM and Hard Disk of a computer system. 3. Data Structures and Algorithms.

MODULE #	**TOPICS**	**Hours**
1	**Introduction:** File Structures: The Heart of the file structure Design, A Short History of File Structure Design, Fundamental File Operations: Physical Files and Logical Files, Opening Files, Closing Files, Reading and Writing, Seeking, Special Characters, The Unix Directory Structure, Physical devices and Logical Files, File-related Header Files, UNIX file System Commands; Secondary Storage : only Disks **Fundamental File Structure Concepts:** Managing Files of Records: Field and Record Organization, Record Access **Text book # 1: Chapter 1 (1.1,1.2), Chapter 2, Chapter 3 (3.1) Chapter 4(4.1.1-4.1.4), Chapter 5 (5.1.1-5.1.4)**	8
	PRACTICAL: 1) File operations: Open, Read, Write and Close 2) Seek operations in Disk and scheduling 3) Seek operations in files 4) Simple Unix fie operations 5) File creation with different fields and records 6) File access methods: sequential and direct	

	RESEARCH: 1) Comparative study of sequential, indexed, and direct file organizations. 2) Design of Hard disks with faster access of files. 3) Comparative analysis of FAT (File Allocation Table), NTFS(New Technology File System), ext4 (Fourth extended File System), APFS (Apple File System), XFS (Extended File System). 4) Dynamic file organization techniques for large-scale databases.	
2	**Organization of Files for Performance**,: Data Compression, Reclaiming Space in files, Internal Sorting and Binary Searching, Key sorting. **Indexing:** What is an Index? A Simple Index for Entry-Sequenced File, Indexes that are too large to hold in Memory, Indexing to provide access by Multiple keys, Retrieval Using Combinations of Secondary Keys, Improving the Secondary Index structure: Inverted Lists, Selective indexes, Binding. **Text book # 1: Chapter 6, Chapter 7(7.1,7.2, 7.5 – 7.10)**	9
	PRACTICAL: 1) File maintenance –addition and deletion of records 2) Sequential searching of records in file 3) Binary searching of records in a file 4) Sorting of files on keys 5) Indexing a file 6) Indexing with multiple keys	
	RESEARCH: 1) Data compression techniques in data files. 2) Performance analysis of sorted files. 3) Efficient storage and retrieval methods for big data files 4) File organization strategies for cloud storage systems. 5) Log-structured file systems and their applications.	
3	**Co-sequential Processing and the Sorting of Large Files**: A Model for Implementing Co-Sequential Processes, Application of the Model to a General Ledger Program, Extension of the Model to include Multiway Merging, A Second Look at Sorting in Memory, Merging as a Way of Sorting Large Files on Disk. **Multi-Level Indexing and B-Trees**: The invention of B-Tree, Statement of the problem, Indexing with Binary Search Trees. **Text book # 1: Chapter 8(8.1 - 8.5.4) , Chapter 9(9.1- 9.3)**	8
	PRACTICAL: 1) Implementation of co-sequential file matching for student and examination records. 2) Merging two sorted files using co-sequential processing. 3) Payroll processing using master and transaction files. 4) Duplicate record detection using co-sequential techniques. 5) Sorting large files stored on SSDs versus HDDs. 6) Indexing with Binary Search Trees.	
	RESEARCH: 1) Parallel co-sequential processing for distributed systems. 2) Machine learning-assisted optimization of co-sequential file matching. 3) Co-sequential processing techniques for big data analytics. 4) GPU-accelerated co-sequential processing methods. 5) Co-sequential processing in cloud-based storage systems.	
4	**Multi-Level Indexing and B -Trees:** B-Trees, Example of Creating a B-Tree, Formal Definition of B-Tree Properties, Worst-case Search Depth,	9

	Deletion, Merging and Redistribution, B* Trees, Buffering of pages; Virtual B-Trees. **Indexed Sequential File Access and Prefix B^+ Trees**: Indexed Sequential Access, maintaining a Sequence Set, adding a Simple Index to the Sequence Set, The Content of the Index: Separators Instead of Keys, The Simple Prefix B^+ Tree and its maintenance, Index Set Block Size, Internal Structure of Index Set Blocks: A Variable-order B- Tree, Loading a Simple Prefix B^+ Trees, B Trees, B^+ Trees and Simple Prefix B^+ Trees in Perspective. **Text book # 1: Chapter 9(9.5 -9.6, 9.10 – 9.12, 9.14 - 9.15.2), Chapter 10**	
	PRACTICAL: 1) Implementation of a B-Tree supporting insertion, deletion, and search operations 2) B-Tree indexing for student record management systems. 3) Implementation of bulk loading in B-Trees. 4) B-Tree-based file indexing system. 5) Implementation of Indexed Sequential Access Method (ISAM). 6) Student database using indexed sequential files. 7) Secondary indexing techniques for sequential files. 8) Multi-level indexing implementation.	
	RESEARCH: 1) Self-tuning B-Tree structures for dynamic workloads. 2) Machine learning-assisted index optimization in B-Trees. 3) Adaptive ISAM structures for big data systems. 4) Cloud-based indexed sequential storage architectures. 5) Hybrid ISAM and B+ Tree architectures.	
5	**Hashing:** Introduction, A Simple Hashing Algorithm, Hashing Functions and Record Distribution, how much Extra Memory should be used? Collision resolution by progressive overflow Buckets, Making deletions, Other collision resolution techniques, Patterns of record access. **Extendible Hashing:** How Extendible Hashing Works, Alternative Approaches. **Text Book # 1: Chapter 11, Chapter 12(12.1, 12.2, 12.6.1-12.6.2.)**	8
	PRACTICAL: 1) Design a file organization using static hashing and Analyse insertion, deletion, and search performance. 2) Implement extendible hashing with directory expansion. 3) Develop a linear hashing file system. 4) Create a file structure using hashing supporting insertion, deletion, and retrieval. 5) Simulate distributed hashing.	
	RESEARCH: 1) Intelligent hash functions to reduce collisions dynamically. 2) Hashing schemes optimized for modern CPU caches. 3) Secure Hash-Based File Structures. 4) Blockchain-Inspired Hash File Systems	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	File Structures-An Object Oriented Approach with C++.	Michael J. Folk, Bill Zoellick, Greg Riccardi	3rd	Pearson Education	2018

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	File Structures using C++	K.R. Venugopal, K.G. Srinivas, P.M. Krishnaraj	-	Tata McGraw-Hill-	2008
2	Database System Concepts	Abraham Silberschatz, Henry F. Korth, S.Sudarshan.	7th	McGraw- Hill	2020
3	Operating System Concepts	Abraham Silberschatz, Peter B. Galvin, Greg Gagne.	4th	Wiley	2018

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**You Tube**	https://www.slideshare.net/shyamujaco/file-structures
2	**NPTEL**	NPTEL Data Storage & Indexing
3	**MIT Open Course Ware**	Operating System Engineering

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Understand the fundamental concepts of file processing operations and storage structures.
CO2	Understand data compression and Indexing in file organization.
CO3	Apply concepts of sorting and merging on multiple files.
CO4	Analyse the sequential and indexing file accessing techniques with appropriate data Structures.
CO5	Illustrate the usage of hashing techniques to organize file structures.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	-	-	-	-	-	-	2	2	-	-
CO2	3	3	2	-	-	-	-	-	-	-	2	2	-	-
CO3	3	3	2	-	-	-	-	-	-	-	2	2	-	-
CO4	3	3	2	-	-	-	-	-	-	-	2	2	-	-
CO5	3	3	2	-	-	-	-	-	-	-	2	2	-	-
TOTAL	15	15	10	-	-	-	-	-	-	-	10	10	-	-
AVG	3	3	2	-	-	-	-	-	-	-	2	2	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. Chalk and Talk with Black Board.
2. ICT based Teaching.
3. Demonstration based Teaching.
4. Simple C programmes to implement algorithms on file operations.

ANALYSIS AND DESIGN OF ALGORITHMS LABORATORY

Course Code	B25CSL46	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	PCCL	**Total SAAE Hours / semester**	02	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. To learn the methods for analyzing algorithms and evaluating their performance. 2. To demonstrate the efficiency of algorithms using asymptotic notations. 3. To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound. 4. To learn the concepts of P and NP complexity classes.

PREREQUISITES
1. Programming in C / Python – Control Structures, Functions & Recursions, Data Types. 2. Data Structures and Applications – Arrays, Stacks, Queue, Linked Lists, Binary Search Tree, Graphs.

Sl. No.	**LIST OF PROGRAMS**
1	Write a C Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.
2	Write a C Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Prim's algorithm.
3	Write C Program to obtain the Topological ordering of vertices in a given digraph using source removal method.
4	Write a C Program to Construct an AVL Tree with 5 elements and perform the necessary rotations for balancing the tree whenever required.
5	Write a C Program to implement Heapsort Algorithm.
6	Write a C Program to solve discrete Knapsack and continuous Knapsack problems using greedy approximation method.
7	Write a C Program to solve All-Pairs Shortest Paths problem using Floyd's algorithm.
8	Write a C Program to find the transitive closure using Warshal's Algorithm.
9	Write C Program to find a subset of a given set S = {sl , s2,.....,sn} of n positive integers whose sum is equal to a given positive integer d.
10	Write C Program for N Queen's problem using Backtracking.

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Apply graph algorithms to solve optimization and connectivity problems using minimum spanning trees, shortest paths, topological sorting, and transitive closure techniques.
CO2	Design and implement efficient data structures and sorting algorithms such as AVL trees and Heap Sort while analyzing their computational efficiency.
CO3	Apply greedy algorithmic strategies to solve optimization problems such as discrete and continuous knapsack problems.
CO4	Develop solutions for combinatorial and constraint satisfaction problems using dynamic programming and backtracking techniques.
CO5	Analyze, test, and evaluate algorithmic solutions with respect to correctness, efficiency, and applicability to real-world computational problems.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO2	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO3	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO4	3	3	3	2	2	-	-	-	1	1	2	3	3	-
CO5	3	3	3	2	2	-	-	-	1	1	2	3	3	-
TOTAL	15	15	15	10	10	-	-	-	5	5	10	15	15	-
AVG	3	3	3	2	2	-	-	-	1	1	2	3	3	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. Chalk and Talk with Black Board.
2. ICT based Teaching.
3. Demonstration based Teaching.

WEB DESIGN

Course Code	B25ISL471	**Total contact hours / week**	02	CIE MARKS	50
Course Category	AEC	**Total SAAE hours / semester**	02	SEE MARKS	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	TOTAL MARKS	100
Total credits	1			Exam Duration	03 Hrs
Mapping to SDG#	4,9&11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable cities & Communities		

COURSE LEARNING OBJECTIVES
1. Design and develop static and dynamic web pages. 2. Familiarize with Client-Side Programming, Server-Side Programming, Active server pages. 3. Learn Database Connectivity to web applications. 4. Infer Object Oriented Programming capabilities of PHP

Sl. No.	**LIST OF PROGRAMS**
1	Write a JavaScript to design a simple calculator to perform the following operations: sum, product, difference and quotient.
2	Write a JavaScript that calculates the squares and cubes of the numbers from 0 to 10and outputs HTML text that displays the resulting values in an HTML table format.
3	Write a JavaScript code that displays text “TEXT-GROWING” with increasing font size in the interval of 100ms in RED COLOR, when the font size reaches 50pt it displays “TEXT-SHRINKING” in BLUE colour. Then the font size decreases to 5pt.
4	Design an XML document to store information about a student in an engineering college affiliated to VTU. The information must include USN, Name, and Name of the College, Branch, Year of Joining, and email id. Make up sample data for 3students. Create a CSS style sheet and use it to display the document
5	Write a PHP program to keep track of the number of visitors visiting the web page and to display this count of visitors, with proper headings.
6	Develop and demonstrate a HTML5 file that includes JavaScript script that uses functions for the following problems: a) Parameter: A string b) Output: The position in the string of the left-most vowel c) Parameter: A number
7	Write a PHP program to display a digital clock which displays the current time ofthe server
8	Write a PHP program to sort the student records which are stored in the database using selection sort.
9	Write the PHP programs to do the following: a. Implement simple calculator operations b. Multiplication of two matrices

TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Fundamentals of Web Development	Richard cotton	3rd	Perason	2022
2	Learning web App development	Semmmy purewal	2nd	Oreilly	2020

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Swayam NPTEL**	https://onlinecourses.swayam2.ac.in/e-learning/preview/nou24_cs09
2	**Geeksforgeeks**	https://www.geeksforgeeks.org/web-tech/web-technology/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Design and develop dynamic web pages with good aesthetic sense of designing and latest technical know-how's.
CO2	Have a good understating of Web Application Terminologies, Internet Tools other web services.
CO3	Learn how to link and publish web sites.
CO4	Appraise the principles of object oriented development using PHP.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	1	3	-	-	-	-	-		2	1	1
CO2	3	2	1	1	3	-	-	-	-	-		2	1	1
CO3	3	2	1	1	3	-	-	-	-	-		2	1	-
CO4	3	2	1	1	3	-	-	-	-	-		2	-	1
TOTAL	12	8	4	4	12	-	-	-	-	-		8	3	3
AVG	3	2	1	1	3	-	-	-	-	-	-	2	1	1

Teaching-Learning Process
Pedagogy (General Instructions):
Teachers can use following strategies to accelerate the attainment of the various course outcomes.
1) Chalk and Talk with Black Board.
2) ICT based Teaching.
3) Demonstration based Teaching.

DATA VISUALIZATION USING PYTHON

Course Code	B25ISL472	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE Hours / semester**	02	**SEE MARKS**	50
L:T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4, 9 and 11	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure **SDG 11:** Sustainable Cities and Communities:		

COURSE LEARNING OBJECTIVES
1. Demonstrate the use of IDLE or PyCharm IDE to create Python Applications. 2. Using Python programming language to develop programs for solving real-world problems. 3. Implementation of Matplotlib for drawing different Plots. 4. Demonstrate working with Seaborn, Bokeh. 5. Working with Plotly for 3D, Time Series and Maps.

PREREQUISITES
1. Basics of programming concept 2. Basics of python

Sl. No.	**LIST OF PROGRAMS**
1	a) Write a Python program that accepts a sentence and find the number of words, digits, uppercase letters and lowercase letters b) Write a Python program to find the string similarity between two given strings
2	a) Write a Python program to Demonstrate how to Draw a Bar Plot using Matplotlib. b) Write a Python program to Demonstrate how to Draw a Scatter Plot using Matplotlib.
3	Write a Python program to Demonstrate how to Draw a Histogram Plot using Matplotlib.
4	Write a Python program to Demonstrate how to Draw a Pie Chart using Matplotlib.
5	Write a Python program to illustrate Linear Plotting using Matplotlib
6	Write a Python program which explains uses of customizing seaborn plots with Aesthetic functions
7	Write a Python program for plotting different types of plots using Bokeh
8	Write a Python program to draw 3D Plots using Plotly Libraries.
9	Write a Python program to draw Time Series using Plotly Libraries
10	Write a Python program for creating Maps using Plotly Libraries.

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Automate the Boring Stuff with Python	Al Sweigart	3rd	No Starch Press	2025
2	Python Programming Using Problem Solving Approach	Reema Thareja	3rd	Oxford University Press	2025

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Think Python: How to Think Like a Computer Scientist	Allen B. Downey	3rd	Green Tea Press	2024
2	Python Data Science Handbook	Jake Vander Plas	3rd	O'REILLY	2023

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**You Tube**	https://www.youtube.com/watch?v=gCCVsvgR2KU
2	**You Tube**	https://www.youtube.com/watch?v=v5MR5JnKcZI
3	**You Tube**	https://www.youtube.com/watch?v=PqFKRqpHrjw
4	**You Tube**	https://www.youtube.com/watch?v=0ZvaDa8eT5s
5	**You Tube**	https://www.youtube.com/watch?v=HZARImviDxg

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Demonstrate the use of IDLE or PyCharm IDE to create Python Applications
CO2	Use Python programming concepts to develop programs for solving real-world problems
CO3	Use Matplotlib for drawing different Plots
CO4	Demonstrate working with Seaborn, Bokeh for visualization.
CO5	Use Plotly for drawing Time Series and Maps.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1	2	2	1	-	-	-	3	1	3	1	2	1
CO2	2	1	3	1	2	-	-	-	2	2	1	2	1	1
CO3	1	3	2	2	1	1	-	-	2	1	1	2	2	1
CO4	3	1	2	2	1	-	-	-	3	1	3	1	2	1
CO5	2	1	3	1	2	-	-	-	2	2	1	2	1	1
TOTAL	11	7	12	8	7	1	-	-	12	7	9	8	8	5
AVG	2.2	1.4	2.4	1.6	1.4	1	-	-	2.4	1.4	1.8	1.6	1.6	1

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. Chalk and Talk
2. ICT based Teaching.
3. Demonstration based Teaching.

TECHNICAL WRITING USING LATEX

Course Code	B25ISL473	**Total contact hours / week**	02	**CIE MARKS**	50
Course Category	AEC	**Total SAAE hours / semester**	02	**SEE MARKS**	50
L: T:P:S	0:0:2:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	4 and 9	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 9:** Industry, Innovation, and Infrastructure		

COURSE LEARNING OBJECTIVES
1.To introduce the basic syntax and semantics of the LaTeX scripting language. 2.To understand the presentation of tables and figures in the document. 3.To illustrate the LaTeX syntax to represent the theorems and mathematical equation. 4.To make use of the libraries (Tikz, algorithm) to design the diagram and algorithms in the document

PREREQUISITES
1.Technical Writing Fundamentals – Understanding of document structure, formatting, and report preparation. 2.Basic Knowledge of Mathematical Notation and Programming Concepts

<table>
<tr><th>Sl. No.</th><th>LIST OF PROGRAMS</th></tr>
<tr><td>1</td><td>Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.</td></tr>
<tr><td>2</td><td>Develop a LaTeX script to create a document that displays the sample Abstract/Summary. 3 Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]</td></tr>
<tr><td>3</td><td>Develop a LaTeX script to create a document that displays the sample Abstract/Summary. 3 Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]</td></tr>
<tr><td>4</td><td>Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry] 5 Develop a LaTeX script to create a document that contains the following table with proper labels</td></tr>
<tr><td>5</td><td>Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry] 5 Develop a LaTeX script to create a document that contains the following table with proper labels
<table>
<tr><th rowspan="2">Sl. No</th><th rowspan="2">USN</th><th rowspan="2">Student Name</th><th colspan="3">Marks</th></tr>
<tr><th>Subject1</th><th>Subject2</th><th>Subject3</th></tr>
<tr><td>1</td><td>1DB25XX001</td><td>Name 1</td><td>89</td><td>60</td><td>90</td></tr>
<tr><td>2</td><td>1DB25XX002</td><td>Name 2</td><td>78</td><td>45</td><td>98</td></tr>
<tr><td>3</td><td>1DB25XX003</td><td>Name 3</td><td>67</td><td>55</td><td>59</td></tr>
</table>
</td></tr>
<tr><td>6</td><td>Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.</td></tr>
</table>

7	a. Develop a LaTeX script to create a simple flowchart using Tikz Library. b. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library
8	Develop a LateX script to plot any Simple Plots/Graph using pgfplots and Tikz Libraries
9	Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.
10	Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section
11	Develop a LaTeX script to create a document that consists of the following two mathematical equation i) (a+b)^2 = a^2 + 2ab + b^2 ii) x = (-b ± √(b² - 4ac)) / 2a iii) φ^λ_σ A_t = Σ_(π∈C_t) sgn(π) φ^λ_σ φ^λ_π
12	Develop a LaTeX script to present an algorithm in the document using algorithm / algorithmic / algorithm2e library.
	Mini- Project (only for Continuous Internal Evaluation)
13	Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice. (Research Paper Preparation, Resume/CV Builder, College Magazine / Newsletter, Question Bank Generator)

REFERENCE BOOK

REFERENCE BOOK #	Book Title	Authors	Edition	Publisher	Year
1	A Guide to LaTeX: Document Preparation for Beginners and Advanced Users	Helmut Kopka and Patrick W. Daly	4th	Addison-Wesley	2003

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Web-Based Interactive Educational Resource**	https://www.overleaf.com/learn
		https://www.latex-project.org/help/documentation/
		https://www.overleaf.com/learn/video-tutorials
		https://ctan.org/

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcomes
CO1	Apply basic LaTeX command to develop simple document.
CO2	Develop LaTeX script to present the tables and figures in the document
CO3	Illustrate LaTeX script to present theorems and mathematical equations in the document.
CO4	Develop programs to generate the complete report with citations and a bibliography.
CO5	Illustrate the use of Tikz and algorithm libraries to design graphics and algorithms in the document

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	-	-	2	-	-	-	-	-	2	3	-	-
CO3	3	3	-	-	2	-	-	-	-	-	2	3	-	-
CO4	3	3	-	-	2	-	-	-	-	-	2	2	-	-
CO5	3	3	-	-	2	-	-	-	-	-	2	2	-	-
TOTAL	15	15	-	-	10	-	-	-	-	-	10	11	-	-
AVG	3	3	-	-	2	-	-	-	-	-	2	2.2	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1. Chalk and Talk
2. ICT based Teaching.
3. Demonstration based Teaching.

BIOLOGY FOR COMPUTER ENGINEERS

Course Code	B25CS48	**Total contact hours / week**	01	**CIE MARKS**	50
Course Category	BSC	**Total SAAE Hours / semester**	16	**SEE MARKS**	50
L:T:P:S	1:0:0:0	**Total Notional Learning Hours**	30	**TOTAL MARKS**	100
Total credits	01			**Exam Duration**	03 Hrs
Mapping to SDG#	3,4,8,13 and 15	**SDG Initiatives**	**SDG 3:** Good Health and Well-being **SDG 4:** Quality Education **SDG 8:** Good Health and Well-being **SDG 13**: Climate Action **SDG 15**: Life on Land		

COURSE LEARNING OBJECTIVES
This course is intended to: 1. To familiarize the students with the basic biological concepts and their engineering applications. 2. To enable the students with an understanding of bio design principles to create novel devices and structures. 3. To provide the students an appreciation of how biological systems can be re-designed as substitute products for natural systems. 4. To motivate the students to develop interdisciplinary vision of biological engineering.

PREREQUISITES
NIL

MODULE #	TOPICS	Hours
1	**Introduction to Biology:** The cell: The basic unit of life, Structure and functions of a cell. Plant Cell and animal cell, Prokaryotic and Eukaryotic cell, Stem cells and their application. Textbook 1: **Chapter 1, Chapter 2** Textbook 5: **Chapter 1, Chapter 2 , Chapter 15**	3
	PRACTICAL: 1) Observation of plant and animal cells under a microscope. 2) Identification of cell organelles using prepared slides/charts. 3) Comparison of prokaryotic and eukaryotic cells.	
	RESEARCH: 1) **Mini Project:** Applications of stem cells in regenerative medicine and healthcare.	
2	**Biomolecules and Biosensors:** Biomolecules: Properties and functions of carbohydrates, proteins, lipids. Short Biosensor History, Biosensor Classification. Textbook 1: **Chapter 3 , Chapter 4** Textbook 3: **Chapter 1**	3
	PRACTICAL: 1) Qualitative tests for carbohydrates, proteins, and lipids. 2) Identification and classification of biosensors based on their components and applications.	

	3) Study of common biosensor models used in healthcare and environmental monitoring.	
	RESEARCH: 1) **Mini Project:** Comparative study of biosensor applications in medical diagnostics, food quality assessment, or environmental monitoring.	
3	**Biochemical Components used in Biosensor Assemblies:** Enzymes, Antibodies, Protein/Peptide Receptors, Nucleic Acids, Whole Cells as Biosensing Elements, Immobilization of Biochemical Elements of Biosensors. Textbook 1: **Chapter 4, Chapter 6** Textbook 3: **Chapter 2**	3
	PRACTICAL: 1) Study of enzymes, antibodies, nucleic acids, and whole cells as biosensing elements. 2) Demonstration of biomolecule immobilization techniques used in biosensors. 3) Identification of biochemical components in selected biosensor assemblies.	
	RESEARCH: 1) **Mini Project:** Investigation of immobilization methods and their impact on biosensor performance.	
4	**Adaptation of Anatomical Principles:** Brain as a CPU system. Eye as a Camera system. Heart as a pump system. Lungs as purification system. Kidney as a filtration system. Textbook 1: **Chapter 10-14**. Textbook 2**: Chapter 7 -18**	3
	PRACTICAL: 1) Study of anatomical systems and their engineering analogies (brain–CPU, eye–camera, heart–pump, lungs–purification, kidney–filtration). 2) Identification of structural and functional similarities between biological and engineering systems. 3) Case study analysis of bio-inspired engineering designs.	
	RESEARCH: 1) **Mini Project:** Bio-inspired engineering applications based on human anatomical systems.	
5	**Biomimetics:** Introduction, Echolocation (ultrasonography / ultrasound Imaging), Photosynthesis (photovoltaic cells, bionic leaf). Birds and insects (flight aerodynamics), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Shark skin (Friction reducing swimsuits), Kingfisher beak (Bullet train). Textbook 1 **Chapter 15-21**, Textbook 4: **Chapter 1, Chapter 9, Chapter 12, Chapter 18**.	3
	PRACTICAL: 1) Study of biomimetic systems inspired by nature. 2) Analysis of biological adaptations and their engineering applications (echolocation, lotus leaf, shark skin, kingfisher beak). 3) Case study of bio-inspired technologies in transportation, energy, and healthcare.	
	1) **RESEARCH:** **Mini Project:** Investigation of a biomimetic innovation and its application in solving an engineering problem.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Biology for Engineers	Rajendra Singh C Rathnakar Rao N, Rajendra Singh C, Rathnakar Rao	--	NPublishing, Bengaluru	2023
2	Human Physiology	Stuart Fox, Krista Rompolski	16th	McGraw-Hill	2022
3	Biosensors: essentials.	Evtugyn, Gennady.	-	Springer, Vol. 84.	2014
4	Biomimetics: Nature-Based Innovation	Yoseph Bar-Cohen	1st	CRC Press.	2012
5	**Biology for Engineers**	Sohini Singh Tanu Allen	--	Navayu Education of India,	2014

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	Wilson and Walker-Principles and Techniques of Biochemistry and Molecular Biology	Andreas Hofmann, Samuel Clokie.	8th	Academic Press	2018
2	3D Bioprinting: Fundamentals, Principles and Applications	Ibrahim Ozbolat	NA	Academic Press	2016.

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Online courses**	https://nptel.ac.in/courses/121106008
2	**Online courses**	https://freevideolectures.com/course/4877/nptel-biology- engineers-other-nonbiologists
3	**Online courses**	https://ocw.mit.edu/courses/20-020-introduction-to-biological-engineering-designspring-2009
4	**Online courses**	https://ocw.mit.edu/courses/20-010j-introduction-to- bioengineering-be-010j-spring2006
5	**Online courses**	https://www.coursera.org/courses?query=biology
6	**Online courses**	https://onlinecourses.nptel.ac.in/noc19_ge31/preview
7	**Online courses**	https://www.classcentral.com/subject/biology
8	**Online courses**	https://www.futurelearn.com/courses/biology-basic-concepts

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Interpret the components of a basic biological cell and their functions
CO2	Understand the principles of bioengineering sensors.
CO3	Compare the adaptation of anatomical principles in day-to-day engineering applications.
CO4	Relate the solution offered by nature to analogous engineering problems.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	1	-	-	1	-	-	1
CO2	3	3	3	3	-	-	-	1	-	-	1	-	-	1
CO3	3	3	3	3	-	-	-	1	-	-	1	-	-	1
CO4	3	3	3	3	-	-	-	1	-	-	1	-	-	1
TOTAL	12	12	12	12	-	-	-	4	-	-	4	-	-	4
AVG	3	3	3	3	-	-	-	1	-	-	1	-	-	1

Teaching-Learning Process

Pedagogy (General Instructions):
These are sample Strategies which teachers can use to accelerate the attainment of the various course outcomes.

1. Explanation via real life problem, situation modelling, and deliberation of solutions, hands-on sessions, reflective and questioning /inquiry-based teaching.
2. Instructions with interactions in classroom lectures (physical/hybrid).
3. Use of ICT tools, including YouTube videos, related MOOCs, AR/VR/MR tools.
4. Flipped classroom sessions (~10% of the classes).
5. Industrial visits, Guests talks and competitions for learning beyond the syllabus.
6. Students' participation through audio-video based content creation for the syllabus (as assignments).
7. Use of gamification tools (in both physical/hybrid classes) for creative learning outcomes.
8. Students' seminars (in solo or group) /oral presentations.

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Group Discussion of Case studies
2. Model Making and seminar/poster presentations

NATIONAL SERVICE SCHEME (NSS)

Course Code	B25NSK391/ 410	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0:2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	3,4,5,6 and 13	**SDG Initiatives**	**SDG 3** - Good Health & well-being. **SDG 4** - Quality Education. **SDG 5** - Gender Equality. **SDG 6** - Clean Water and sanitation. **SDG 13** - Climate Action.		

COURSE LEARNING OBJECTIVES

1. Understand the community in general in which they work.
2. Identify the needs and problems of the community and involve them in problem –solving.
3. Develop among themselves a sense of social & civic responsibility & utilize their knowledge in finding practical solutions to individual and community problems.
4. Develop competence required for group-living and sharing of responsibilities & gain skills in mobilizing community participation to acquire leadership qualities and democratic attitudes.
   Develop capacity to meet emergencies and natural disasters & practice national integration and social harmony in general.

PREREQUISITES

1 Students should have service oriented mind-set and social concern.

2 Students should have dedication to work at any remote place, any time with available resources and proper time management for the other works.

3 Students should be ready to sacrifice some of the timely will and wishes to achieve service Oriented targets on time.

GENERAL INSTRUCTIONS - PEDAGOGY

These are sample strategies; which teachers can use to accelerate the attainment of the various course outcomes.

In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the activities will develop students' theoretical and applied social and cultural skills.

State the need for NSS activities and its present relevance in the society and Provide real-life examples.

Support and guide the students for self-planned activities.

You will also be responsible for assigning homework, grading assignments and quizzes, and documenting students' progress in real activities in the field.

Encourage the students for group work to improve their creative and analytical skills.

Sl. #	CONTENTS
1	Organic farming, Indian Agriculture (Past, Present and Future) Connectivity for marketing.
2	Waste management– Public, Private and Govt organization, 5 R's.
3	Setting of the information imparting club for women leading to contribution in social and economic issues.
4	Water conservation techniques – Role of different stakeholders– Implementation.
5	Preparing an actionable business proposal for enhancing the village income and approach for implementation.
6	Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education.
7	Developing Sustainable Water management system for rural areas and implementation approaches.
8	Contribution to any national level initiative of Government of India. For eg. Digital India, Skill India, Swatch Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc.
9	Spreading public awareness under rural outreach programs. (minimum one program).
10	Social connect and responsibilities.
11	Plantation and adoption of plants. Know your plants.
12	Organize National integration and social harmony events /workshops /seminars. (Minimum 02 programs).
13	Govt. school Rejuvenation and helping them to achieve good infrastructure.

NOTE:

1. Student/s in individual or in a group should select any two activities in the beginning of the semester. Should work on the selected activities till the end of the semester for successful completion as per the instructions of NSS officer with the consent of HOD of the department.
2. At the end of semester, activity report should be submitted for evaluation.

Semester	Sl. #	TOPICS TO BE COVERED
3rd Sem/4th Sem B25NSK391/410	1	Waste management– Public, Private and Govt organization, 5 R's.
	2	Water conservation techniques – Role of different stakeholders– Implementation.
	3	Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education.
	4	Contribution to any national level initiative of Government of India. For eg. Digital India, Skill India, Swatch Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc.
	5	Plantation and adoption of plants. Know your plants.

Suggested Learning Resources
1) NSS Course Manual, Published by NSS Cell, VTU Belagavi.
2) Government of Karnataka, NSS cell, activities reports and its manual.
3) Government of India, NSS cell, Activities reports and its manual.

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Understand the importance of his / her responsibilities towards society.
CO2	Analyse the environmental and societal problems/issues and will be able to design solutions for the same.
CO3	Evaluate the existing system and to propose practical solutions for the same for sustainable development.
CO4	Implement government or self-driven projects effectively in the field.
CO5	Develop capacity to meet emergencies and natural disasters & practice national integration and social harmony in general.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	-	-	3	-	-	3	-	3	-	-	3	-	-	3
CO2	-	-	3	-	-	3	-	3	-	-	3	-	-	3
CO3	-	-	3	-	-	3	-	3	-	-	3	-	-	3
CO4	-	-	3	-	-	3	-	3	-	-	3	-	-	3
CO5	-	-	3	-	-	3	-	3	-	-	3	-	-	3
TOTAL	-	-	15	-	-	15	-	15	-	-	15	-	-	15
AVG	-	-	3			3		3			3			3

SL. #	Practice Session Description
1	Student Presentation - Selection of topics
2	Commencement of activity and its progress
3	Execution of Activity
4	Case study based Assessment, Individual performance
5	Sector wise study and its consolidation
6	Video based seminar for 10 minutes by each student at the end of semester with Report.

Assessment Details for CIE

Weightage	CIE – 100	
Presentation - Selection of topic.	20 Marks	Implementation strategies of the project (NSS work). 1) The report should be evaluated by Department NSS Coordinator. 2) It should be signed by Department NSS Coordinator, NSS Program Officer and HOD of the department.
Commencement of activity and its progress	20 Marks	
Case study based Assessment Individual performance	20 Marks	
Sector wise study and its consolidation	20 Marks	
Video based seminar for 10 minutes by each student at the end of semester with report.	20 Marks	
Total marks for the course in any one semester	**100 Marks**	

PHYSICAL EDUCATION (PE) (SPORTS AND ATHLETICS)

Course Code	B25PEK391/ 410	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0:2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	3, 5 and 8	**SDG Initiatives**	**SDG 3:** Good Health and Well-Being. **SDG 5:** Gender Equality. **SDG 8:** Decent Work and Economic Growth.		

COURSE LEARNING OBJECTIVES
1. To provide hands-on experience in selected games and sports management. 2. To create awareness of fitness among the student's.

PREREQUISITES
NIL

MODULE #	TOPICS	Hours
1	Orientation A) Lifestyle B) Health & Wellness	2
2	General Fitness & Components of Fitness A) Warming up (Free Hand exercises) B) Strength – Push-up / Pull-ups	2
3	Specific games (Any one to be selected by the student) A) Kabaddi – Hand touch, Toe Touch, Thigh Hold, Ankle hold and Bonus. B) Kho-Kho – Giving Kho, Single Chain, Pole dive, Pole turning, 3-6 Up.	2
4	Ethics and Moral Values A) Ethics in Sports B) Moral Values in Sports and Games	2
5	Specific Games (Any one to be selected by the student) A) Volleyball – Attack, Block, Service, Upper Hand Pass and Lower Hand Pass. B) Athletics (Track Events) – Any event as per availability of Ground.	2

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1.	Health , Exercise and Fitness	Muller, J P	-	Sports Publication	2018
2.	Track and field Marking and Athletics Officiating Manual	M J Vishwanath	-	Silver Star Shimoga	2002
3	Fitness and Wellness	Werner W K. Hoeger & Sharon	9th	Morton Publishing Company, Canada	1990
4.	Health Fitness Instructors handbook	Howley. ET & Franks BD	3rd	Human Kinetics, Champaign Illinois	1997

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**You Tube**	https://www.youtube.com/watch?v=dczxtSt1Kjk https://youtube.com/watch?v=TXGejZBgNeo
2	**E-books**	https://books.google.co.in/books/about/Athlete_s_Guide_to_Career_Planning.html?id=lUp5QgAACAAJ&redir_esc=y https://drbgrpublications.in/wp-content/uploads/books/Dr.BGR-ISBN-46.pdf

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Understand the fundamental concepts and skills of Physical Education, Health, Nutrition and Fitness.
CO2	Familiarization of health-related Exercises, Sports for overall growth and development.
CO3	Understand and practice of Traditional Games.
CO4	Understand the ethics and moral values in sports and athletics
CO5	Understand the roles and responsibilities of organisation and administration of sports and games.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO2	-	-	-	-	-	-	3	3	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	2	-	-	-	-	-	3
CO5	-	-	-	-	-	-	2	2	-	-	-	-	-	2
TOTAL	-	-	-	-	-	-	5	11	-	-	-	-	-	5
AVG	-	-	-	-	-	-	2.5	2.2	-	-	-	-	-	2.5

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use following strategies to accelerate the attainment of the various course outcomes.

1) Activity Based Learning Participation.
2) Quiz, Presentation
3) Assignments & Models.

YOGA

Course Code	B25YOK391/ B25YOK410	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0: 2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	3,4,12,16	**SDG Initiatives**	**SDG3:** Good Health and Well-Being **SDG4:** Quality Education **SDG12:** Responsible Consumption and Production **SDG16:** Peace and Justice Strong Institutions		

COURSE LEARNING OBJECTIVES

1. Understand the philosophy, objectives, and importance of Yoga in daily life.
2. Practice Yogic exercises and Suryanamaskara for maintaining physical and mental health.
3. Perform different types of Asanas with proper posture, breathing techniques, and safety precautions.
4. Develop discipline, concentration, flexibility, emotional stability, and stress management through regular Yogic practices.
5. Adopt Yoga as a lifelong practice for promoting a healthy lifestyle and overall well-being.

PREREQUISITES

1) Students should have an interest in maintaining physical and mental fitness.
2) Basic awareness of health, hygiene, and wellness is desirable.
3) Students should be willing to participate actively in Yoga practice sessions.
4) Comfortable clothing suitable for Yoga practice is recommended.
5) Students should follow safety precautions and instructor guidelines during practice.

GENERAL INSTRUCTIONS - PEDAGOGY

The following teaching strategies may be adopted to achieve the course outcomes.

1) Explain the importance and philosophy of Yoga using interactive lectures.
2) Demonstrate all Yogic practices before student participation.
3) Encourage regular practice of Yoga techniques.
4) Explain the benefits and precautions of every Asana.
5) Conduct practical demonstrations individually and in groups.
6) Encourage students to maintain a Yoga practice diary.
7) Assess students through continuous observation and practical performance.
8) Promote awareness regarding healthy lifestyle and stress management.
9) Organize Yoga demonstrations, awareness programs, and group activities.
10) Motivate students to practice Yoga regularly beyond classroom hours.

Sl. #	CONTENTS
1	Introduction to Yoga
2	Meaning and Definition of Yoga
3	Aim and Objectives of Yoga
4	Importance of Yoga in Daily Life
5	Prayer and its Importance
6	Different Schools of Yoga
7	Rules and Regulations for Yoga Practice
8	Misconceptions about Yoga
9	Yogic Practices for Common People
10	Positive Health through Yoga
11	Suryanamaskara Prayer
12	Meaning and Benefits of Suryanamaskara
13	Suryanamaskara – 12 Counts (2 Rounds)
14	Sitting Asanas – Padmasana
15	Sitting Asanas – Vajrasana
16	Standing Asanas – Vrikshasana
17	Standing Asanas – Trikonasana
18	Prone Asanas – Bhujangasana
19	Prone Asanas – Shalabhasana
20	Supine Asanas – Utthitadvipadasana
21	Supine Asanas – Ardha Halasana
22	Technique, Benefits and Precautions of each Asana

Suggested Learning Resources
Text Books 1) B.K.S. Iyengar – Light on Yoga 2) Swami Satyananda Saraswati – Asana Pranayama Mudra Bandha 3) Swami Vivekananda – Raja Yoga 4) Ministry of AYUSH – Common Yoga Protocol **Reference Books** 1) Hatha Yoga Pradipika 2) Patanjali Yoga Sutras 3) Yoga for Health – Swami Kuvalayananda **Online Resources** 1) Ministry of AYUSH Yoga Portal 2) Common Yoga Protocol Manual 3) NPTEL Yoga and Wellness Lectures 4) WHO Physical Activity Guidelines

COURSE OUTCOMES:

After successful completion of the course, students will be able to:

CO #	Course Outcome Statement
CO1	Explain the philosophy, objectives, and significance of Yoga for healthy living.
CO2	Demonstrate Suryanamaskara and basic Yogic practices correctly.
CO3	Perform various sitting, standing, prone, and supine Asanas with proper techniques and precautions.
CO4	Apply Yoga practices for improving physical fitness, mental well-being, and stress management.
CO5	Develop lifelong healthy habits through regular Yoga practice and self-discipline.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	-	-	3	-	-	3	-	3	-	-	3	-	1	-
CO2	-	-	3	-	-	3	-	3	-	-	3	-	1	-
CO3	-	-	3	-	-	3	-	3	-	-	3	-	1	-
CO4	-	-	3	-	-	3	-	-	-	-	3	-	1	-
CO5	-	-	3	-	-	3	-	3	-	-	3	-	1	-
TOTAL	-	-	15	-	-	15	-	15	-	-	15	-	5	-
AVG	-	-	3	-	-	3	-	3.75	-	-	3	-	1	-

SL. #	Practice Session Description
1	Introduction to Yoga and Prayer
2	Demonstration of Suryanamaskara
3	Practice of Sitting Asanas
4	Practice of Standing Asanas
5	Practice of Prone Asanas
6	Practice of Supine Asanas
7	Individual Assessment of Asanas
8	Group Yoga Demonstration
9	Benefits and Precautions Discussion
10	Final Practical Evaluation

Assessment Details for CIE

Weightage	CIE – 100	Implementation Strategies for Yoga Practical Evaluation 1) The Yoga Practical Record and practical performance shall be evaluated by the Course Handling Faculty / Yoga Instructor. 2) The Yoga Practical Record shall be signed by the Course Handling Faculty / Yoga Instructor and the Head of the Department (HOD).
Attendance and Active Participation in Yoga Sessions	20 Marks	
Demonstration of Suryanamaskara	20 Marks	
Performance of Asanas (Technique, Posture and Breathing)	20 Marks	
Viva Voce (Benefits, Precautions and Applications of Yoga)	20 Marks	
Final Practical Demonstration and Yoga Practice Record Submission	20 Marks	
Total marks for the course in any one semester	**100 Marks**	

MUSIC

Course Code	B25MUK391/ 410	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0:2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	3,4 and 11	**SDG Initiatives**	**SDG 3:** Good Health & well-being. **SDG 4:** Quality Education **SDG 11:** Sustainable Cities and Communities		

COURSE LEARNING OBJECTIVES
1. Identify the major traditions of Indian music, both through notations and aurally. 2. Analyze compositions with respect to musical and lyrical content. 3. Demonstrate an ability to use music technology appropriately in a variety of settings.

PREREQUISITES
Nil

MODULE #	TOPICS	Hours
1	Preamble: Contents of the curriculum intend to promote music as a language to develop analytical, creative, and intuitive understanding. Students must experience music through study and direct participation in improvisation and composition. ORIGIN OF INDIAN MUSIC: Evolution of the Indian music system understanding of: Shruthi, Nada, Swara, Laya, Raga, Tala, Mela.	6
	PRACTICAL: 1) Identification and demonstration of Shruthi, Swara and Laya. 2) Listening exercises on different Ragas and Talas.	
	RESEARCH: 1) Study the evolution of Indian Music from Vedic to modern periods. 2) Compare Carnatic and Hindustani music systems.	
2	Compositions: Introduction to the types of composition in Carnatic Music: Geethe, Jathi , Swara , Swarajathi , Varna, Krithi , Thillana , Notation System.	6
	PRACTICAL: 1) Practice singing basic Geethes. 2) Notation reading and writing exercises	
	RESEARCH: 1) Study the characteristics of Geethe, Swarajathi, and Krithi. 2) Analyze the structure of Carnatic music compositions.	

3	Composers: Biography and contributions of Purandaradasa, Thyagaraja, Mysore Vasudevacharya.	6
	PRACTICAL: 1) Listening and appreciation of compositions by renowned composers. 2) Presentation on the contributions of selected composers.	
	RESEARCH: 1) Study the life and works of Purandaradasa. 2) Compare the compositional styles of Thyagaraja and Mysore Vasudevacharya.	
4	Music Instruments: Classification and Construction of String instruments , Wind instruments, Percussion instruments , Idiophones (Ghana Vaadya) , Examples of each class of instruments.	5
	PRACTICAL: 1) Identification of different categories of musical instruments. 2) Demonstration of string, wind, and percussion instruments.	
	RESEARCH: 1) Study the classification of Indian musical instruments. 2) Analyze the role of instruments in classical music performances.	
5	Abhyasa Gana: Singing the swara exercises (Sarale Varase only), Notation writing for Sarale Varase , Suladi Saptha Tala (only in Mayamalavagowla Raga) , Singing 4 Geethe in Malahari , One Jathi Swara, One Krithi in a Mela Raga	5
	PRACTICAL: 1) Singing Sarale Varase and Geethes in prescribed Ragas. 2) Practice of Jathi Swara and Krithi performance.	
	RESEARCH: 1) Study the importance of Abhyasa Gana in Carnatic music training. 2) Analyze the significance of Tala and Raga in music learning.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Theory of Music	Vidushi Vasantha Madhavi	-	Prism Publication	2007
2	Karnataka Sangeetha Darpana –Vol. I	T. Sachidevi & T. Sharadha (Thirumalai Sisters)	-	Shreenivasa Prakashana	2018

REFERENCE BOOKS

Ref Book #	Book Title	Authors	Edition	Publisher	Year
1	Classical Music of India: A Practical Guide	Lakshminarayana Subramaniam & Viji Subramaniam	-	-	2018

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO 1	Discuss the Indian system of music and relate it to other genres (Cognitive Domain).
CO 2	Experience the emotions of the composer and develop empathy (Affective Domain).
CO 3	Respond to queries on various patterns in a composition (Psychomotor Domain).

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	-	-	-	-	-	-	-	-	-	-	-	1
CO2	-	-	-	-	-	1	-	2	2	1	-	-	-	1
CO3	1	2	-	-	1	-	-	-	2	2	-	-	-	1
TOTAL	3	3	-	-	1	1	-	2	4	3	-	-	-	3
AVG	1.5	1.5	-	-	1	1	-	2	2	1.5	-	-	-	1

Teaching-Learning Process

Pedagogy (General Instructions):

Teachers can use the following strategies to accelerate the attainment of the various course outcomes.

1) Classroom Teaching
2) ICT – PowerPoint Presentation
3) Audio & Video Visualization Tools

FINE ARTS

Course Code	B25FAK391/410	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0:2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	4,5,8, 11 and 12	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 5:** Gender Equality **SDG 8:** Decent Work and Economic Growth **SDG 11:** Sustainable Cities and Communities **SDG 12:** Responsible Consumption and Production		

COURSE LEARNING OBJECTIVES
1. Understand the use of art materials, tools, and mediums to execute precise drawings, paintings, or sculptures. 2. Cultivate innovation by experimenting with various artistic styles and finding your own unique artistic voice. 3. Analyze, evaluate, and interpret the content, form, and historical context of both masterworks and contemporary art.

PREREQUISITES
Observational Skills, Creative Potential:, Familiarity with basic design principles (line, shape, form, color) and general proficiency in standard sketching or digital media is highly beneficial

MODULE #	TOPICS	Hours
1	**Foundations of Visual Arts:** **The 7 Elements of Art:** Learn the building blocks of visual communication—Line, Shape, Space, Value, Form, Texture, and Color. **The Principles of Design:** Understand composition and how to guide the viewer's eye using balance, rhythm, emphasis, proportion, and harmony.	3
	PRACTICAL: 1) Recreate a single, famous historical painting using only a limited-color palette or a single pencil.	
	RESEARCH: 1) Choose one contemporary digital artwork or traditional masterpiece. Write a brief 300-word critique explicitly outlining how the artist manipulated Colour and Value to direct the viewer's eye across the painting's Balance and Harmony.	

2	**Practical Drawing & Sketching:** **Basic Shapes & Perspective:** Learn to draw basic geometric forms (cubes, spheres, cylinders) while applying light, shadow, and vanishing points. **Still-Life Observation:** Set up a simple still life (e.g., a book and a fruit). Focus on measuring proportions and capturing realistic textures.	3
	PRACTICAL: 1) Draw the same basic geometric forms three times using three distinct lighting scenarios (e.g., direct overhead light, directional side light, and diffused ambient light).	
	RESEARCH: 1) Research the Renaissance chiaroscuro technique (e.g., Caravaggio or Rembrandt). Analyze how historical masters manipulated light and shadow to create dramatic, three-dimensional realism, and write a brief analysis of how they used core forms to construct their compositions.	
3	**Painting Techniques & Colour Theory**: **The Colour Wheel:** Explore the relationships between primary, secondary, and tertiary colours. Learn how to mix tints, shades, and tones. **Medium Exploration:** Apply colour theory by painting a simple landscape or object in acrylics or watercolours. Practice techniques like washing, blending, and layering.	3
	PRACTICAL: 1)Before painting your final landscape, create a 5-step gradient scale (from pure hue to white for tints, and black for shades). Practice matching the warmth/coolness of the light (e.g., warm afternoon light vs. cool morning mist) by slightly altering your colour mixtures.	
	RESEARCH: 1)Research how specific colour combinations create spatial illusions. Investigate how artists use simultaneous contrast and complementary palettes to make foreground objects pop against a background.	
4	**History & Appreciation of Art:** **Historical Eras Overview:** A brief look at major global art movements, from ancient cave paintings and civilizations (like the Indus Valley) to the Renaissance and Modern Art] **Discussion & Critique:** Analyze famous historical and contemporary masterpieces and understand how to evaluate and discuss art critically	3
	PRACTICAL: 1)Visit the National Gallery of Modern Art (NGMA) on Palace Road in Bengaluru. Select one Indian modern or contemporary piece (e.g., works by Raja Ravi Varma, Amrita Sher-Gil, or M.F. Husain). Sketch or visually map out the structural composition, colour palette, and focal points of the artwork.	
	RESEARCH: 1)Trace the evolution of a specific artistic medium across different historical eras, such as the transition from ancient Indus Valley/Mesopotamian terracotta and bronze crafting to Renaissance fresco/tempera painting.	

5	**Portfolio Building & Next Steps:** **Curating Your Work:** Learn how to photograph, mount, and present your studio pieces. **Resources & Advice:** Set up a personal digital art journal or blog.	2
	PRACTICAL: 1) Build your personal blog or digital art journal using portfolio-focused platforms like Squarespace or WordPress. Implement a dedicated "Process" or "Work-in-Progress" category to document your sketches, failures, and material research, transforming your journal into an engaging case study rather than just an image gallery.	
	RESEARCH: 1) Analyze competitor and professional artist websites. Research user experience (UX) trends to decide whether a grid-based gallery or a minimalist "one artwork per page" layout best suits your specific medium and tells the story of your artistic growth.	

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Drawing on the Right Side of the Brain	**Betty Edwards**	4th	Tarcher Perigee	2012
2	Colour Mixing Recipes	**William F. Powell**	SPI	Walter Foster Publishing	2004
3	Art Fundamentals: Theory and Practice	**Otto G. Ocvirk**	11th	McGraw-Hill Education	2009

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**Website for free digital painting**	https://www.ctrlpaint.com/
2	**Online Course**	https://www.proko.com/browse/lessons
3	**Video**	https://www.youtube.com/watch?v=Iw7HxsR_CO0
4	**Video**	https://www.youtube.com/watch?v=EaZyPAjdvSo
5	**NPTEL Course**	https://onlinecourses.nptel.ac.in/e-learning/preview/noc20_hs09
6	**NPTEL Course**	https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_hs48
7	**NPTEL Course**	https://onlinecourses.swayam2.ac.in/e-learning/ preview/ cec22_as04

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	**Identify and apply** the foundational elements of art and principles of design to effectively produce visual compositions.
CO2	**Execute** technical skills in drawing, sketching, and painting using a variety of mediums (e.g., acrylics, water colours) while demonstrating an understanding of colour theory and perspective.
CO3	**Analyze and critique** historical and contemporary artworks, recognizing major global art movements and cultural contexts.
CO4	**Curate and present** a digital portfolio of personal studio work, utilizing professional presentation and documentation techniques.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	1	1	-	-	-	-	-	-	-	-	-	1
CO2	2	2	1	1	-	-	-	-	-	-	-	-	-	1
CO3	2	2	1	1	-	-	-	-	-	-	-	-	-	1
CO4	2	2	1	1	-	-	-	-	-	-	-	-	-	1
TOTAL	8	8	4	4	-	-	-	-	-	-	-	-	-	4
AVG	2	2	1	1	-	-	-	-	-	-	-	-	-	1

Teaching-Learning Process

Pedagogy (General Instructions):

Hands-on Sessions / Guided Practice, Demonstrations, Interactive Problem Solving / Q&A Sessions and Project-Based Learning

SUSTAINABLE DEVELOPMENT GOALS

Course Code	B25SDG391	**Total contact hours / week**	-	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	28	**SEE MARKS**	-
L:T:P:S	0:0:0:2	**Total Notional Learning Hours**	28	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	-
Mapping to SDG#	1 to 17	**SDG Initiatives**	**SDG 1:** No Poverty **SDG 2:** Zero Hunger **SDG 3:** Good Health and Well Being **SDG 4:** Quality Education **SDG 5:** Gender Equality **SDG 6:** Clean Water and Sanitation **SDG 7:** Affordable and Clean Energy **SDG 8:** Decent Work and Economic Growth **SDG 9:** Industry, Innovation, and Infrastructure **SDG 10:** Reduced Inequalities **SDG 11:** Sustainable Cities and Communities **SDG 12:** Responsible Consumption and Production **SDG 13:** Climate Action **SDG 14:** Life below Water **SDG 15:** Life on Land **SDG 16:** Peace, Justice and Strong Institutions **SDG 17:** Partnerships for the Goals		

COURSE LEARNING OBJECTIVES
1. Identifying where SDGs align with disciplinary learning 2. Developing sustainability related learning activities 3. Designing transformative and applied learning experiences 4. Supporting students to become active contributors to sustainable development

PREREQUISITES
NIL

MODULE #	**TOPICS**	**Hours**
1	The Millennium Development Goals MDGs	01
2	From MDGs to SDGs: What are the Sustainable Development Goals?	01
3	Pillars of Sustainable Development Goals	01
4	Sustainable Development Goals	01

URL's:

1) https://glorew.com
2) https://www.good-deeds-day.org
3) https://sdgs.un.org/goals
4) https://undp.org
5) https://www.niti.gov.in/divisions/division/sustainable-development-goal
6) https://www.mospi.gov.in

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Explain the meaning and significance of Sustainable Development Goals
CO2	Critically analyse the meaning and importance of education in the context of Sustainable Development
CO3	Understand the holistic framework built around five pillars

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**
CO2	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**
CO3	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**
TOTAL	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**
AVG	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**

ADDITIONAL MATHEMATICS II

Common to all branches

Course Code	B25DIP411	**Total contact hours / week**	02	**CIE MARKS**	100
Course Category	NCMC	**Total SAAE Hours / semester**	20	**SEE MARKS**	-
L:T:P:S	0:0:0:0	**Total Notional Learning Hours**	48	**TOTAL MARKS**	100
Total credits	0			**Exam Duration**	3 hours
Mapping to SDG#	4,8 and 9	**SDG Initiatives**	**SDG 4:** Quality Education **SDG 8**: Decent work and Economic Growth **SDG 9**: Industry, Innovation, and Infrastructure		

COURSE LEARNING OBJECTIVES
1. Familiarize the importance of vector calculus and linear algebra in engineering. 2. Have an insight into solving ordinary differential equations by using Laplace transform techniques.

PRE-REQUISITES
1. Basics of differentiation and vectors.

MODULE #	TOPICS	Hours
1	**VECTOR ALGEBRA** Scalar and vectors, addition and subtraction of vectors, Dot and Cross products-problems. Scalar triple product-problems. **Text book 1: Chapter 3(3.1-3.6)**	5
2	**VECTOR DIFFERENTIATION** Differentiation of vector functions. Velocity and acceleration of a particle moving on a space curve. Gradient, Divergence, Curl-simple problems. Solenoidal and irrotational vector fields- problems. **Text book 1: Chapter 8(8.1- 8.5)**	6
3	**HIGHER-ORDER DIFFERENTIAL EQUATIONS** Linear homogeneous and non-homogeneous differential equations of second and higher-order with constant coefficients. Solution by using the inverse differential operator method. [Particular Integrals restricted to $R(x) = e^{ax}$, sinax, cosax]. **Text book 1: Chapter 13(13.4-13.8)**	6
4	**LINEAR ALGEBRA** Introduction. Rank of a matrix by elementary row operations, consistency and solution of system of linear equations, Gauss elimination method. Eigen values and eigen vectors of a square matrix using Rayleigh's power method-Problems. **Text book 1: Chapter 2(2.4-2.8,2.13)**	5
5	**LAPLACE TRANSFORMS** Definitions. Laplace transforms of elementary functions. Properties. Inverse Laplace transforms. Applications of Laplace transforms to solve differential equations. **Text book 1: Chapter 21 (21.2-21.4, 21.7-21.13, 21.15)**	6

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Authors	Edition	Publisher	Year
1	Higher Engineering mathematics	**B.S Grewal**	43rd	Khanna Publishers,	2015
2	Higher Engineering Mathematics	**H.K Dass**	10th	John Wiley and sons	2016

REFERENCE BOOKS

Reference Book #	Book Title	Authors	Edition	Publisher	Year
1	A Text book of Engineering Mathematics	N.P.Bali and Manish Goyal	10th	Laxmi Publications	2022
2	Linear Algebra and its Applications	David C Lay	4th	Pearson Publisher	2018
3	Engineering Mathematics - II	Pandurangappa C	2nd	Sanguine Technical Publications	2022

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	**NPTEL**	https://nptel.ac.in/courses/111105160
2	**NPTEL**	https://nptel.ac.in/courses/127106019
3	**MOOCs**	https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-spring-2019/
4	**MOOCs**	https://ocw.mit.edu/courses/18-330-introduction-to-numerical-analysis-spring- 2012/pages/syllabus/
5	**VTU e-learning**	http://elearning.vtu.ac.in

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Solve simple problems on vector algebra.
CO2	Evaluate gradient , divergence and curl and extend their application to solenoidal and irrotational vectors.
CO3	Solve differential equations of second and higher order with constant coefficients.
CO4	Solve the system of linear equations and compute eigenvalues and eigenvectors.
CO5	Apply Laplace transform techniques for time domain and solving differential equations.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO2	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO3	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO4	3	3	-	2	-	-	-	2	2	-	2	2	-	-
CO5	3	3	-	2	-	-	-	2	2	-	2	2	-	-
TOTAL	15	15	-	10	-	-	-	10	10	-	10	10	-	-
AVG	3	3	-	2	-	-	-	2	2	-	2	2	-	-

Teaching-Learning Process

Pedagogy (General Instructions):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the delivered lessons shall develop students' theoretical and applied mathematical skills.
2. State the need for Mathematics with Engineering Studies and Provide real-life examples.
3. Support and guide the students for self–study.
4. You will also be responsible for assigning homework, grading assignments and quizzes, and documenting students' progress.
5. Encourage the students to group learning to improve their creative and analytical skills.
6. Show short-related video lectures in the following ways:
    a. As an introduction to new topics (pre-lecture activity).
    b. As a revision of topics (post-lecture activity).
    c. As additional examples (post-lecture activity).
    d. As an additional material of challenging topics (pre-and post-lecture activity)
    e. As a model solution of some exercises (post-lecture activity).

READY2RISE

Course Code	B25RRK412	Total contact hours / week	02	CIE MARKS	100
Course Category	NCMC	Total SAAE Hours / semester	14	SEE MARKS	-
L:T:P:S	1:0:0:1	Total Notional Learning Hours	28	TOTAL MARKS	100
Total credits	0			Exam Duration	-
Mapping to SDG#	4	SDG Initiatives	SDG 4: Quality Education		

COURSE LEARNING OBJECTIVES

1. Understand industry expectations and placement processes.
2. Develop professional communication and workplace behaviour.
3. Strengthen analytical, logical, and problem-solving abilities.
4. Enhance teamwork, leadership, and professional networking skills.
5. Create a career roadmap aligned with employability goals.

PREREQUISITES

1. Enrolment in the 4th Semester of a B.E./B.Tech. program.
2. Basic understanding of engineering domains and career opportunities.
3. Awareness of common job roles related to their branch.
4. Fundamental analytical and logical reasoning skills.
5. Ability to approach problems systematically.
6. Willingness to engage in self-assessment activities.

Module #	Topic	Mode of Delivery	Hours
1	Self-Awareness and Personality Assessment	Classroom & Activity	4
2	Communication Skills for Employability	Classroom & Activity	4
3	Emotional Intelligence and Interpersonal Skills	Classroom & Activity	4
4	Teamwork and Leadership Development	Classroom & Activity	4
5	Professional Excellence and Workplace Readiness	Classroom & Activity	4
6	Placement Readiness Foundation	Classroom & Activity	4
		Total	**24**

PRESCRIBED TEXT BOOKS

Text Book #	Book Title	Author(s)	Edition	Publisher	Year
1	Effective Technical Communication	M. Ashraf Rizvi	2nd	McGraw Hill Education	2017
2	Professional Communication	Aruna Koneru	2nd	McGraw Hill Education	2016

REFERENCE BOOKS

Reference Book #	Book Title	Author(s)	Edition	Publisher	Year
1	What Color Is Your Parachute?	Richard N. Bolles	2021 Edition (50th Anniversary Edition)	Ten Speed Press	2021
2	Designing Your Life	Bill Burnett & Dave Evans	Updated Edition	Knopf (Penguin Random House)	2016
3	The Start-Up of You	Reid Hoffman & Ben Casnocha	Revised Edition	Crown Business	2012

E-Resources: E-books / Online Course materials / Online Courses / Video Lectures:

Sl. #	Type of E-Resource	URL
1	British Council Learn English	British Council Learn English
2	Coursera Communication Skills Courses	Coursera Communication Skills Courses
3	Toastmasters International	Toastmasters International

COURSE OUTCOMES:

At the end of the Course, student will be able to:

CO #	Course Outcome Statement
CO1	Explain the campus recruitment process, industry expectations, and employability requirements.
CO2	Demonstrate professional communication, presentation, and interpersonal skills.
CO3	Apply logical reasoning, analytical thinking, and structured problem-solving techniques.
CO4	Exhibit teamwork, leadership, professional ethics, and workplace etiquette.
CO5	Develop a personalized career action plan and demonstrate readiness for future placement activities.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	-	-	-	-	-	-	2	2	-	-	-	3
CO2	-	-	-	-	-	-	-	1	3	3	-	-	-	2
CO3	2	3	1	-	-	-	-	-	1	1	-	-	-	2
CO4	-	1	-	-	-	1	-	3	3	2	2	-	-	2
CO5	1	1	-	-	-	-	-	2	2	3	1	-	-	3
TOTAL	5	6	1	-	-	1	-	6	11	11	3	-	-	12
AVG	1.6	1.5	1	-	-	1	-	2	2.2	2.2	1.5	-	-	2.4

Teaching-Learning Process

Pedagogy (General Instructions):

1. Career Awareness and Industry Expectations
   - Interactive sessions on career pathways and emerging opportunities
   - Industry trend analysis and discussions
   - Guest lectures by industry professionals and alumni
2. Professional Communication Excellence
   - Communication skill workshops
   - Demonstrations and guided practice
   - Role plays and situational conversations
3. Critical Thinking and Problem Solving
   - Problem-Based Learning (PBL)
   - Case study analysis
   - Brainstorming and inquiry-based discussions
4. Teamwork, Leadership and Workplace Skills
   - Collaborative learning and team projects
   - Leadership simulations
   - Team-building activities
5. Professional Branding and Career Planning
   - Workshop-based learning
   - Portfolio and resume development sessions
   - Career coaching and mentoring
6. Placement Foundation Readiness
   - Practice-oriented training
   - Mock aptitude tests
   - Mock interviews and feedback sessions

Non Credit Mandatory Course (NCMC)
Assessment Details (both CIE and SEE):
CIE: Each question carrying 1 Mark (No Negative Marking for wrong answers)
CIE-1: 50 Multiple Choice Questions – 60 Minutes - 50X1=50 Marks
CIE-2: 50 Multiple Choice Questions – 60 Minutes - 50X1=50 Marks
TOTAL= 100 Marks
SEE: Not Applicable